

FINANCE &
RISK
ANALYTICS
PART A

Business Report

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1 - Define the problem and perform Exploratory Data Analysis:

-Problem Statement

Context.

In the realm of modern finance, businesses encounter the perpetual challenge of managing debt obligations effectively to maintain a favorable credit standing and foster sustainable growth. Investors keenly scrutinize companies capable of navigating financial complexities while ensuring stability and profitability. A pivotal instrument in this evaluation process is the balance sheet, which provides a comprehensive overview of a company's assets, liabilities, and shareholder equity, offering insights into its financial health and operational efficiency. In this context, leveraging available financial data, particularly from preceding fiscal periods, becomes imperative for informed decision-making and strategic planning.

Objective.

A renowned credit rating organization wants to develop a Financial Health Assessment Tool. With the help of the tool, it endeavors to empower businesses and investors with a robust mechanism for evaluating the financial well-being and creditworthiness of companies. By harnessing machine learning techniques, the organization aims to analyze historical financial statements and extract pertinent insights to facilitate informed decision-making via the tool. Specifically, the organization foresees facilitating the following with the help of the tool:

1. Debt Management Analysis: Identify patterns and trends in debt management practices to assess the ability of businesses to fulfill financial obligations promptly and efficiently, and identify potential cases of default.
2. Credit Risk Evaluation: Evaluate credit risk exposure by analyzing liquidity ratios, debt-to-equity ratios, and other key financial indicators to ascertain the likelihood of default and inform investment decisions.

As a part of the data science team in the organization, you have been provided with the financial metrics of different companies. The task is to analyze the data provided and develop a predictive model leveraging machine learning techniques to identify whether a given company will default on its debt repayments in the next two quarters. The predictive model will help the organization anticipate potential challenges with the financial performance of the companies and enable proactive risk mitigation strategies.

Data Overview.

>Top 5 Rows of the dataset;

	Co_Code	Co_Name	_Operating_Expense_Rate	_Research_and_development_expense_rate	_Cash_flow_rate	_Interest_bearing_debt_interest_rate	_Tax_rate_A	_Cash_Flow
0	16974	Hind.Cables	8.820000e+09	0.000000e+00	0.462045	0.000352	0.001417	
1	21214	Tata Tele. Mah.	9.380000e+09	4.230000e+09	0.460116	0.000716	0.000000	
2	14852	ABG Shipyard	3.800000e+09	8.150000e+08	0.449893	0.000496	0.000000	
3	2439	GTL	6.440000e+09	0.000000e+00	0.462731	0.000592	0.009313	
4	23505	Bharati Defence	3.680000e+09	0.000000e+00	0.463117	0.000782	0.400243	

Table1- Top 5 Rows of the dataset

>last 5 Rows of the dataset;

	Co_Code	Co_Name	_Operating_Expense_Rate	_Research_and_development_expense_rate	_Cash_flow_rate	_Interest_bearing_debt_interest_rate	_Tax_rate_A	_Cash
2053	2743	Kothari Ferment.	3.021580e-04	6.490000e+09	0.477066	0.000000	0.183014	
2054	21216	Firstobj.Tech.	1.371450e-04	0.000000e+00	0.465211	0.000658	0.000000	
2055	142	Diamines & Chem.	2.114990e-04	8.370000e+09	0.480248	0.000502	0.000000	
2056	18014	IL&FS Engg.	3.750000e+09	0.000000e+00	0.474670	0.000578	0.306205	
2057	43229	Channel Nine	2.981110e-04	0.000000e+00	0.467203	0.000826	0.000000	

Table2- Last 5 Rows of the dataset

>Shape of the dataset;

There are 2058 Rows and 58 Columns present in the dataset.

>Basic Info of the dataset;

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 2058 entries, 0 to 2057

Data columns (total 58 columns):

#	Column	Non-Null Count	Dtype
0	Co_Code	2058 non-null	int64
1	Co_Name	2058 non-null	object
2	_Operating_Expense_Rate	2058 non-null	float64
3	_Research_and_development_expense_rate	2058 non-null	float64
4	_Cash_flow_rate	2058 non-null	float64
5	_Interest_bearing_debt_interest_rate	2058 non-null	float64
6	_Tax_rate_A	2058 non-null	float64
7	_Cash_Flow_Per_Share	1891 non-null	float64
8	_Per_Share_Net_profit_before_tax_Yuan_	2058 non-null	float64
9	_Realized_Sales_Gross_Profit_Growth_Rate	2058 non-null	float64
10	_Operating_Profit_Growth_Rate	2058 non-null	float64
11	_Continuous_Net_Profit_Growth_Rate	2058 non-null	float64
12	_Total_Asset_Growth_Rate	2058 non-null	float64
13	_Net_Value_Growth_Rate	2058 non-null	float64
14	_Total_Asset_Return_Growth_Rate_Ratio	2058 non-null	float64
15	_Cash_Reinvestement_perc	2058 non-null	float64
16	_Current_Ratio	2058 non-null	float64
17	_Quick_Ratio	2058 non-null	float64
18	_Interest_Expense_Ratio	2058 non-null	float64
19	_Total_debt_to_Total_net_worth	2037 non-null	float64
20	_Long_term_fund_suitability_ratio_A	2058 non-null	float64
21	_Net_profit_before_tax_to_Paid_in_capital	2058 non-null	float64
22	_Total_Asset_Turnover	2058 non-null	float64
23	_Accounts_Receivable_Turnover	2058 non-null	float64
24	_Average_Collection_Days	2058 non-null	float64
25	_Inventory_Turnover_Rate_times	2058 non-null	float64
26	_Fixed_Assets_Turnover_Frequency	2058 non-null	float64
27	_Net_Worth_Turnover_Rate_times	2058 non-null	float64
28	_Operating_profit_per_person	2058 non-null	float64
29	_Allocation_rate_per_person	2058 non-null	float64
30	_Quick_Assets_to_Total_Assets	2058 non-null	float64
31	_Cash_to_Total_Assets	1962 non-null	float64
32	_Quick_Assets_to_Current_Liability	2058 non-null	float64
33	_Cash_to_Current_Liability	2058 non-null	float64
34	_Operating_Funds_to_Liability	2058 non-null	float64
35	_Inventory_to_Working_Capital	2058 non-null	float64
36	_Inventory_to_Current_Liability	2058 non-null	float64
37	_Long_term_Liability_to_Current_Assets	2058 non-null	float64
38	_Retained_Earnings_to_Total_Assets	2058 non-null	float64
39	_Total_income_to_Total_expense	2058 non-null	float64
40	_Total_expense_to_Assets	2058 non-null	float64
41	_Current_Asset_Turnover_Rate	2058 non-null	float64
42	_Quick_Asset_Turnover_Rate	2058 non-null	float64
43	_Cash_Turnover_Rate	2058 non-null	float64
44	_Fixed_Assets_to_Assets	2058 non-null	float64
45	_Cash_Flow_to_Total_Assets	2058 non-null	float64
46	_Cash_Flow_to_Liability	2058 non-null	float64
47	_CFO_to_Assets	2058 non-null	float64
48	_Cash_Flow_to_Equity	2058 non-null	float64
49	_Current_Liability_to_Current_Assets	2044 non-null	float64
50	_Liability_Assets_Flag	2058 non-null	int64
51	_Total_assets_to_GNP_price	2058 non-null	float64
52	_No_credit_Interval	2058 non-null	float64
53	_Degree_of_Financial_Leverage_DFL	2058 non-null	float64
54	_Interest_Coverage_Ratio_Interest_expense_to_EBIT	2058 non-null	float64
55	_Net_Income_Flag	2058 non-null	int64
56	_Equity_to_Liability	2058 non-null	float64
57	Default	2058 non-null	int64

dtypes: float64(53), int64(4), object(1)

memory usage: 932.7+ KB

Table3- Basic Info of the dataset

-Observations: The dataset has 53 float, 4 int and 1 object datatype present.

>Basic Info after remove the ‘_’ (startwith) from all the column;

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 2058 entries, 0 to 2057

Data columns (total 58 columns):

#	Column	Non-Null Count	Dtype
0	Co_Code	2058 non-null	int64
1	Co_Name	2058 non-null	object
2	Operating_Expense_Rate	2058 non-null	float64
3	Research_and_development_expense_rate	2058 non-null	float64
4	Cash_flow_rate	2058 non-null	float64
5	Interest_bearing_debt_interest_rate	2058 non-null	float64
6	Tax_rate_A	2058 non-null	float64
7	Cash_Flow_Per_Share	1891 non-null	float64
8	Per_Share_Net_profit_before_tax_Yuan_	2058 non-null	float64
9	Realized_Sales_Gross_Profit_Growth_Rate	2058 non-null	float64
10	Operating_Profit_Growth_Rate	2058 non-null	float64
11	Continuous_Net_Profit_Growth_Rate	2058 non-null	float64
12	Total_Asset_Growth_Rate	2058 non-null	float64
13	Net_Value_Growth_Rate	2058 non-null	float64
14	Total_Asset_Return_Growth_Rate_Ratio	2058 non-null	float64
15	Cash_Reinvestment_perc	2058 non-null	float64
16	Current_Ratio	2058 non-null	float64
17	Quick_Ratio	2058 non-null	float64
18	Interest_Expense_Ratio	2058 non-null	float64
19	Total_debt_to_Total_net_worth	2037 non-null	float64
20	Long_term_fund_suitability_ratio_A	2058 non-null	float64
21	Net_profit_before_tax_to_Paid_in_capital	2058 non-null	float64
22	Total_Asset_Turnover	2058 non-null	float64
23	Accounts_Receivable_Turnover	2058 non-null	float64
24	Average_Collection_Days	2058 non-null	float64
25	Inventory_Turnover_Rate_times	2058 non-null	float64
26	Fixed_Assets_Turnover_Frequency	2058 non-null	float64
27	Net_Worth_Turnover_Rate_times	2058 non-null	float64
28	Operating_profit_per_person	2058 non-null	float64
29	Allocation_rate_per_person	2058 non-null	float64
30	Quick_Assets_to_Total_Assets	2058 non-null	float64
31	Cash_to_Total_Assets	1962 non-null	float64
32	Quick_Assets_to_Current_Liability	2058 non-null	float64
33	Cash_to_Current_Liability	2058 non-null	float64
34	Operating_Funds_to_Liability	2058 non-null	float64
35	Inventory_to_Working_Capital	2058 non-null	float64
36	Inventory_to_Current_Liability	2058 non-null	float64
37	Long_term_Liability_to_Current_Assets	2058 non-null	float64
38	Retained_Earnings_to_Total_Assets	2058 non-null	float64
39	Total_income_to_Total_expense	2058 non-null	float64
40	Total_expense_to_Assets	2058 non-null	float64
41	Current_Asset_Turnover_Rate	2058 non-null	float64
42	Quick_Asset_Turnover_Rate	2058 non-null	float64
43	Cash_Turnover_Rate	2058 non-null	float64
44	Fixed_Assets_to_Assets	2058 non-null	float64
45	Cash_Flow_to_Total_Assets	2058 non-null	float64
46	Cash_Flow_to_Liability	2058 non-null	float64
47	CFO_to_Assets	2058 non-null	float64
48	Cash_Flow_to_Equity	2058 non-null	float64
49	Current_Liability_to_Current_Assets	2044 non-null	float64
50	Liability_Assets_Flag	2058 non-null	int64
51	Total_assets_to_GNP_price	2058 non-null	float64
52	No_credit_Interval	2058 non-null	float64
53	Degree_of_Financial_Leverage_DFL	2058 non-null	float64
54	Interest_Coverage_Ratio_Interest_expense_to_EBIT	2058 non-null	float64
55	Net_Income_Flag	2058 non-null	int64
56	Equity_to_Liability	2058 non-null	float64
57	Default	2058 non-null	int64

dtypes: float64(53), int64(4), object(1)

memory usage: 932.7+ KB

Table4- Basic info after remove ‘_’

>Checking for no. of unique values;

Co_Code	2058
Co_Name	2058
Operating_Expense_Rate	1495
Research_and_development_expense_rate	629
Cash_flow_rate	1888
Interest_bearing_debt_interest_rate	813
Tax_rate_A	985
Cash_Flow_Per_Share	900
Per_Share_Net_profit_before_tax_Yuan_	876
Realized_Sales_Gross_Profit_Growth_Rate	1939
Operating_Profit_Growth_Rate	2015
Continuous_Net_Profit_Growth_Rate	2014
Total_Asset_Growth_Rate	922
Net_Value_Growth_Rate	1757
Total_Asset_Return_Growth_Rate_Ratio	1428
Cash_Reinvestment_perc	1690
Current_Ratio	1972
Quick_Ratio	1970
Interest_Expense_Ratio	1716
Total_debt_to_Total_net_worth	1949
Long_term_fund_suitability_ratio_A	2014
Net_profit_before_tax_to_Paid_in_capital	1798
Total_Asset_Turnover	283
Accounts_Receivable_Turnover	1109
Average_Collection_Days	1935
Inventory_Turnover_Rate_times	1151
Fixed_Assets_Turnover_Frequency	1079
Net_Worth_Turnover_Rate_times	529
Operating_profit_per_person	1484
Allocation_rate_per_person	2051
Quick_Assets_to_Total_Assets	2058
Cash_to_Total_Assets	1962
Quick_Assets_to_Current_Liability	2058
Cash_to_Current_Liability	2056
Operating_Funds_to_Liability	2058
Inventory_to_Working_Capital	1931
Inventory_to_Current_Liability	1932
Long_term_Liability_to_Current_Assets	1398
Retained_Earnings_to_Total_Assets	2058
Total_income_to_Total_expense	2056
Total_expense_to_Assets	2058
Current_Asset_Turnover_Rate	1973
Quick_Asset_Turnover_Rate	1743
Cash_Turnover_Rate	1440
Fixed_Assets_to_Assets	2054
Cash_Flow_to_Total_Assets	2058
Cash_Flow_to_Liability	2058
CFO_to_Assets	2058
Cash_Flow_to_Equity	2058
Current_Liability_to_Current_Assets	2044
Liability_Assets_Flag	2
Total_assets_to_GNP_price	2058
No_credit_Interval	2057
Degree_of_Financial_Leverage_DFL	1940
Interest_Coverage_Ratio_Interest_expense_to_EBIT	1945
Net_Income_Flag	1
Equity_to_Liability	2058
Default	2

dtype: int64

Table5- Unique Values of the dataset

>Statistical Summary of the dataset;

	count	mean	std	min	25%	50%	75%	max
Co_Code	2058.0	1.757211e+04	2.189289e+04	4.000000	3.674000e+03	6.240000e+03	2.428075e+04	7.249300e+04
Operating_Expense_Rate	2058.0	2.052389e+09	3.252624e+09	0.000100	1.578727e-04	3.330330e-04	4.110000e+09	9.980000e+09
Research_and_development_expense_rate	2058.0	1.208634e+09	2.144568e+09	0.000000	0.000000e+00	1.994130e-04	1.550000e+09	9.980000e+09
Cash_flow_rate	2058.0	4.652426e-01	2.266269e-02	0.000000	4.600991e-01	4.634450e-01	4.680691e-01	1.000000e+00
Interest_bearing_debt_interest_rate	2058.0	1.113022e+07	9.042595e+07	0.000000	2.760280e-04	4.540450e-04	6.630660e-04	9.900000e+08
Tax_rate_A	2058.0	1.147770e-01	1.524457e-01	0.000000	0.000000e+00	3.709890e-02	2.161909e-01	9.996963e-01
Cash_Flow_Per_Share	1891.0	3.199856e-01	1.529979e-02	0.169449	3.149890e-01	3.206479e-01	3.259178e-01	4.622268e-01
Per_Share_Net_profit_before_tax_Yuan_	2058.0	1.769673e-01	3.015730e-02	0.000000	1.666039e-01	1.756421e-01	1.858854e-01	7.923477e-01
Realized_Sales_Gross_Profit_Growth_Rate	2058.0	2.276117e-02	2.170104e-02	0.004282	2.205831e-02	2.210001e-02	2.215200e-02	1.000000e+00
Operating_Profit_Growth_Rate	2058.0	8.481083e-01	4.589093e-03	0.736430	8.479740e-01	8.480386e-01	8.481147e-01	1.000000e+00
Continuous_Net_Profit_Growth_Rate	2058.0	2.173915e-01	5.678779e-03	0.000000	2.175741e-01	2.175961e-01	2.176198e-01	2.332046e-01
Total_Asset_Growth_Rate	2058.0	5.287663e+09	2.912615e+09	0.000000	4.315000e+09	6.225000e+09	7.220000e+09	9.980000e+09
Net_Value_Growth_Rate	2058.0	5.189504e+06	2.077918e+08	0.000000	4.362833e-04	4.554170e-04	4.883758e-04	9.330000e+09
Total_Asset_Return_Growth_Rate_Ratio	2058.0	2.641004e-01	2.415661e-03	0.251620	2.637383e-01	2.640161e-01	2.643097e-01	3.586288e-01
Cash_Reinvestment_perc	2058.0	3.771970e-01	2.737311e-02	0.025828	3.707295e-01	3.789678e-01	3.855575e-01	1.000000e+00
Current_Ratio	2058.0	1.336249e+06	6.061917e+07	0.000000	6.567062e-03	8.945370e-03	1.350542e-02	2.750000e+09
Quick_Ratio	2058.0	2.775510e+07	4.448654e+08	0.000000	2.946399e-03	5.284241e-03	8.902983e-03	9.230000e+09
Interest_Expense_Ratio	2058.0	6.312913e-01	6.785512e-03	0.525126	6.306116e-01	6.307999e-01	6.317437e-01	8.121652e-01
Total_debt_to_Total_net_worth	2037.0	1.071429e+07	2.696960e+08	0.000000	3.924894e-03	7.270721e-03	1.306869e-02	9.940000e+09
Long_term_fund_suitability_ratio_A	2058.0	8.973310e-03	3.485186e-02	0.004129	5.162031e-03	5.517000e-03	6.415301e-03	1.000000e+00
Net_profit_before_tax_to_Paid_in_capital	2058.0	1.753994e-01	2.622348e-02	0.000000	1.658623e-01	1.745683e-01	1.844450e-01	7.921047e-01
Total_Asset_Turnover	2058.0	1.286405e-01	1.006216e-01	0.000000	6.146927e-02	1.034483e-01	1.679160e-01	9.190405e-01
Accounts_Receivable_Turnover	2058.0	4.159864e+07	5.047673e+08	0.000000	7.446260e-04	1.081432e-03	1.854463e-03	9.740000e+09
Average_Collection_Days	2058.0	2.629786e+07	4.109967e+08	0.000000	3.576384e-03	6.001272e-03	8.638997e-03	8.800000e+09
Inventory_Turnover_Rate_times	2058.0	2.030227e+09	3.077250e+09	0.000000	1.909297e-04	1.910000e+07	3.815000e+09	9.990000e+09
Fixed_Assets_Turnover_Frequency	2058.0	1.230898e+09	2.649289e+09	0.000000	2.278950e-04	5.995245e-04	8.423224e-03	9.990000e+09
Net_Worth_Turnover_Rate_times	2058.0	3.957710e-02	4.239591e-02	0.008871	2.048387e-02	2.870968e-02	4.435484e-02	1.000000e+00
Operating_profit_per_person	2058.0	4.036693e-01	5.358970e-02	0.000000	3.913864e-01	3.950781e-01	4.008927e-01	1.000000e+00
Allocation_rate_per_person	2058.0	5.725559e+06	1.979500e+08	0.000000	4.671612e-03	1.062969e-02	2.457491e-02	8.280000e+09
Quick_Assets_to_Total_Assets	2058.0	3.421979e-01	2.103925e-01	0.000000	1.734827e-01	3.061276e-01	4.845435e-01	9.889440e-01
Cash_to_Total_Assets	1962.0	7.993675e-02	9.862260e-02	0.000184	2.061909e-02	4.563187e-02	9.771301e-02	9.250180e-01
Quick_Assets_to_Current_Liability	2058.0	1.190476e+07	3.122923e+08	0.000000	3.616304e-03	5.972976e-03	9.608533e-03	8.820000e+09
Cash_to_Current_Liability	2058.0	9.282507e+07	7.851899e+08	0.000101	1.085476e-03	2.684338e-03	7.540535e-03	9.170000e+09
Operating_Funds_to_Liability	2058.0	3.482338e-01	3.840302e-02	0.026274	3.377032e-01	3.450257e-01	3.541402e-01	1.000000e+00
Inventory_to_Working_Capital	2058.0	2.777491e-01	1.844394e-02	0.000000	2.770093e-01	2.772511e-01	2.777111e-01	1.000000e+00
Inventory_to_Current_Liability	2058.0	5.786346e+07	6.278795e+08	0.000000	2.890842e-03	6.781166e-03	1.275116e-02	9.600000e+09
Long_term_Liability_to_Current_Assets	2058.0	7.340107e+07	6.693526e+08	0.000000	0.000000e+00	2.587130e-03	1.049684e-02	9.310000e+09
Retained_Earnings_to_Total_Assets	2058.0	9.303546e-01	2.976067e-02	0.000000	9.278868e-01	9.350756e-01	9.409371e-01	9.727688e-01
Total_income_to_Total_expense	2058.0	2.357977e-03	4.644258e-04	0.000000	2.186964e-03	2.297452e-03	2.433146e-03	1.028413e-02
Total_expense_to_Assets	2058.0	3.109208e-02	3.870042e-02	0.000853	1.270426e-02	2.086322e-02	3.530120e-02	1.000000e+00
Current_Asset_Turnover_Rate	2058.0	1.273303e+09	2.839741e+09	0.000000	1.504698e-04	2.461660e-04	1.264005e-03	9.990000e+09
Quick_Asset_Turnover_Rate	2058.0	2.571768e+09	3.453544e+09	0.000000	1.511758e-04	3.794085e-04	5.790000e+09	1.000000e+10

Table6- Statistical Summary Of the dataset

Total_expense_to_Assets	2058.0	3.109208e-02	3.870042e-02	0.000853	1.270426e-02	2.086322e-02	3.530120e-02	1.000000e+00
Current_Asset_Turnover_Rate	2058.0	1.273303e+09	2.839741e+09	0.000000	1.504698e-04	2.461660e-04	1.264005e-03	9.990000e+09
Quick_Asset_Turnover_Rate	2058.0	2.571768e+09	3.453544e+09	0.000000	1.511758e-04	3.794085e-04	5.790000e+09	1.000000e+10
Cash_Turnover_Rate	2058.0	2.653696e+09	2.821245e+09	0.000100	1.737418e-03	1.730000e+09	4.550000e+09	9.990000e+09
Fixed_Assets_to_Assets	2058.0	4.042760e+06	1.834006e+08	0.000000	9.650577e-02	2.138107e-01	4.150287e-01	8.320000e+09
Cash_Flow_to_Total_Assets	2058.0	6.442325e-01	4.505929e-02	0.000000	6.333645e-01	6.432462e-01	6.541577e-01	1.000000e+00
Cash_Flow_to_Liability	2058.0	4.599747e-01	3.288112e-02	0.032583	4.574802e-01	4.593408e-01	4.617433e-01	9.051198e-01
CFO_to_Assets	2058.0	5.797344e-01	6.375060e-02	0.000000	5.503790e-01	5.825431e-01	6.123215e-01	9.751973e-01
Cash_Flow_to_Equity	2058.0	3.146292e-01	1.277967e-02	0.000000	3.127830e-01	3.146423e-01	3.165460e-01	5.692307e-01
Current_Liability_to_Current_Assets	2044.0	3.935178e-02	4.797815e-02	0.000000	2.177539e-02	3.265229e-02	4.394684e-02	1.000000e+00
Liability_Assets_Flag	2058.0	3.401361e-03	5.823606e-02	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00
Total_assets_to_GNP_price	2058.0	2.779397e+07	4.717714e+08	0.000000	9.124052e-04	2.479550e-03	7.004449e-03	9.820000e+09
No_credit_Interval	2058.0	6.236856e-01	1.163052e-02	0.408682	6.233274e-01	6.237496e-01	6.240452e-01	9.563871e-01
Degree_of_Financial_Leverage_DFL	2058.0	2.785248e-02	1.383854e-02	0.012845	2.677558e-02	2.681466e-02	2.702943e-02	4.643880e-01
Interest_Coverage_Ratio_Interest_expense_to_EBIT	2058.0	5.654355e-01	1.153538e-02	0.172065	5.651580e-01	5.653149e-01	5.662324e-01	6.667613e-01
Net_Income_Flag	2058.0	1.000000e+00	0.000000e+00	1.000000	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
Equity_to_Liability	2058.0	4.252852e-02	5.952518e-02	0.003946	2.040787e-02	2.846004e-02	4.343255e-02	1.000000e+00
Default	2058.0	1.068999e-01	3.090610e-01	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00

Table6- Statistical Summary Of the dataset

-Observations:

- **Co_Code:** High average (17,572.11) and wide variation (std. dev. 21,892.89), indicating some companies have very low values.
- **Operating Expense Rate:** Average is high (2.05 billion), with some companies having very high expenses (max 9.98 billion).
- **R&D Expense Rate:** Average is 1.21 billion, but many companies invest little or nothing in R&D.
- **Cash Flow Rate:** Average (0.465) suggests strong cash flow, but some companies have cash flow challenges (min 0).
- **Tax Rate:** Average tax rate is about 11.48%, with some firms facing very high tax liabilities.
- **Growth Rates:**
- **Operating Profit Growth Rate:** High average (0.848) indicates consistent profit growth.
- **Continuous Net Profit Growth Rate:** Moderate growth (0.217).
- **Asset Growth:** Extremely high average (5.29 billion), suggesting aggressive growth strategies, but may raise sustainability concerns.
- **Liquidity Ratios:** Both Current and Quick Ratios are high, indicating strong liquidity, but some companies may have unusually high assets compared to liabilities.
- **Debt Metrics:** High average debt levels suggest some firms may be over-leveraged. Interest coverage ratio (0.565) shows moderate ability to meet interest payments, but some may be at risk.
- **Cash Metrics:** High average cash turnover (2.65 billion) indicates good cash generation, but variability points to differences in efficiency.
- **Profitability:** Average net profit relative to capital (0.175) shows reasonable profitability.
- **Risk Indicators:** A default rate of 10.69% indicates some companies are at risk of financial instability.
- We can see that Co_Code and Co_Name are not relevant for this exercise.
- So we will drop these variables.

Exploratory Data Analysis.

>Univariate Analysis;

In the Default column there are only unique values are present.

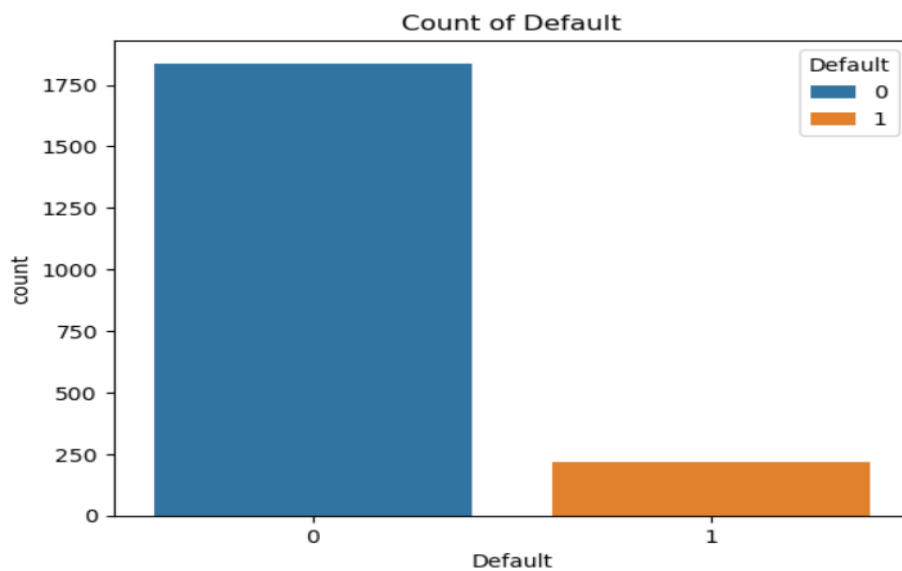


Figure 1

-Observations: In the Default column there are only 2 unique values are present. We can see from the above figure that there non defaulter is larger than the defaulter it suggests that a majority of the companies in the dataset are financially stable and not at risk of defaulting. And the percentage of the defaulter is 10.689990281827017.

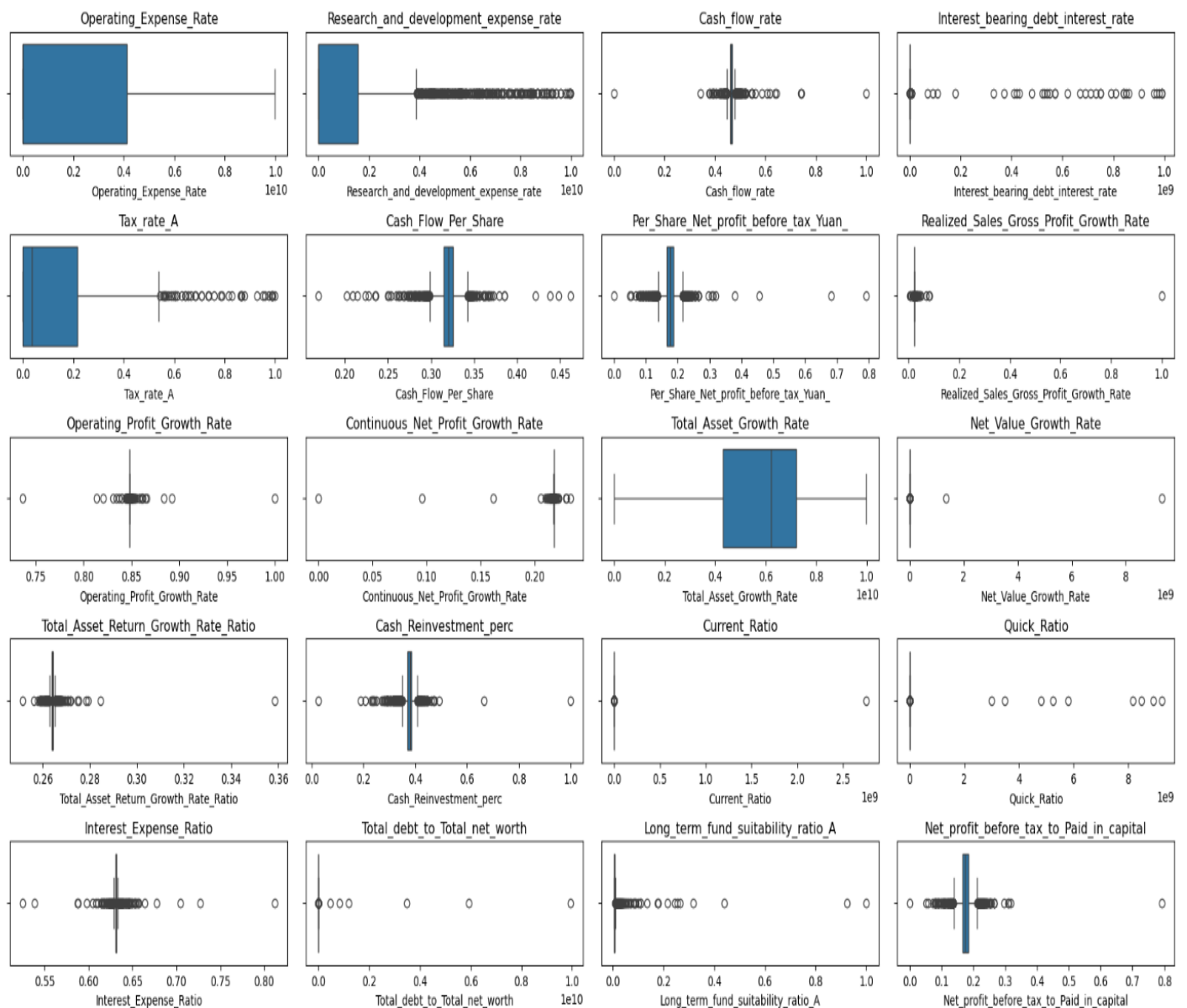


Figure 2

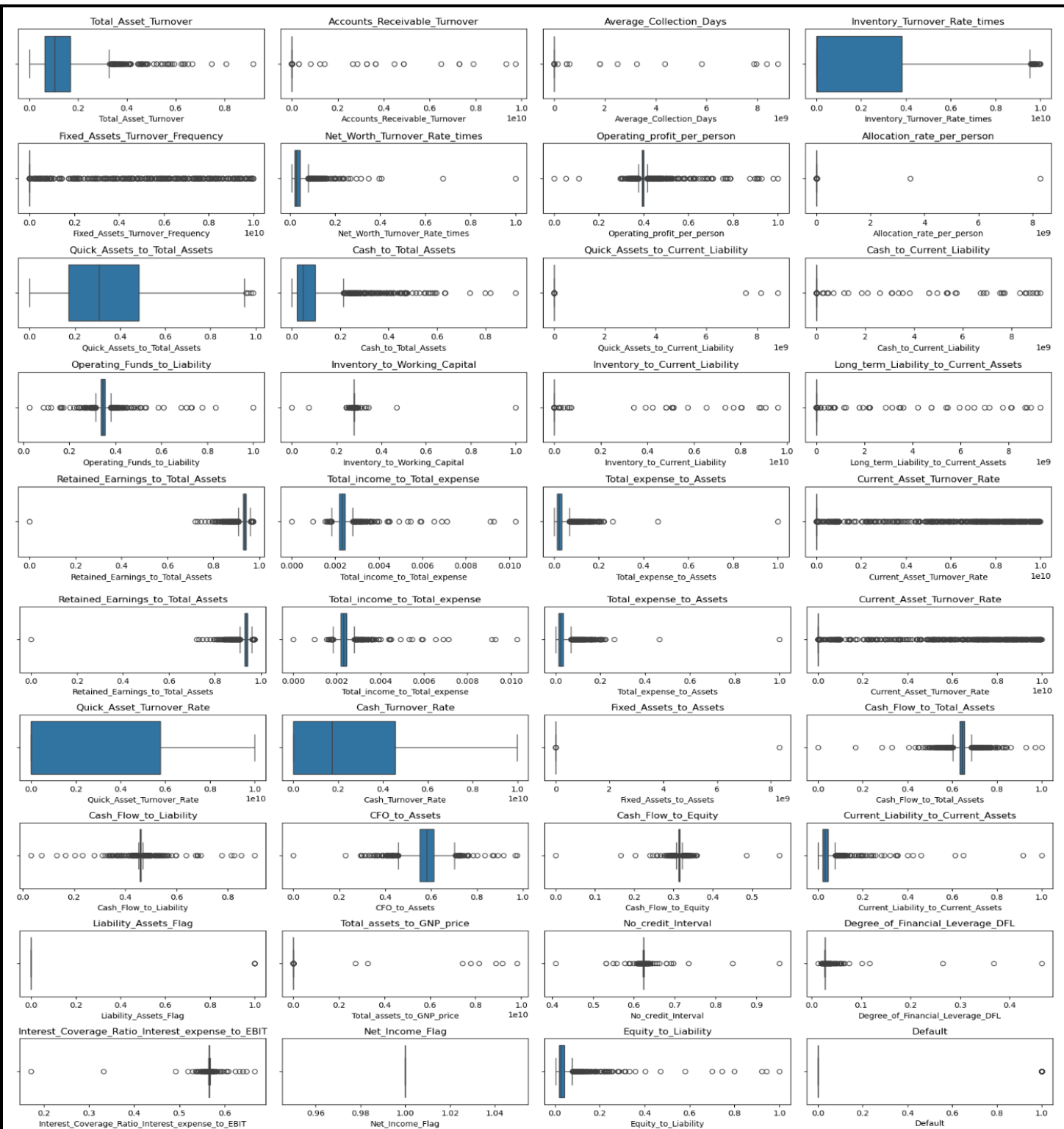


Figure 2

-Observations:

- Each boxplot shows the median, quartiles, and potential outliers of the data.
- The presence of points outside the whiskers indicates potential outliers. Approximately all the numerical variables showed outlier. Only the operating expense rate, total asset growth rate, quick asset turnover rate and cash turnover rate shows no outlier.
- The position of the median line within the box can indicate skewness. Most of the box plot shows positively skewed.
- The length of the box and the distance between the whiskers are provide variability into the dataset.
- There are multiple numerical columns are present, so it can compare their boxplots and reveal differences in central tendency and variability across variables.
- Some of the box plots shows non-normality because of extreme outlier or long tail and that is important for certain statistical analysis.

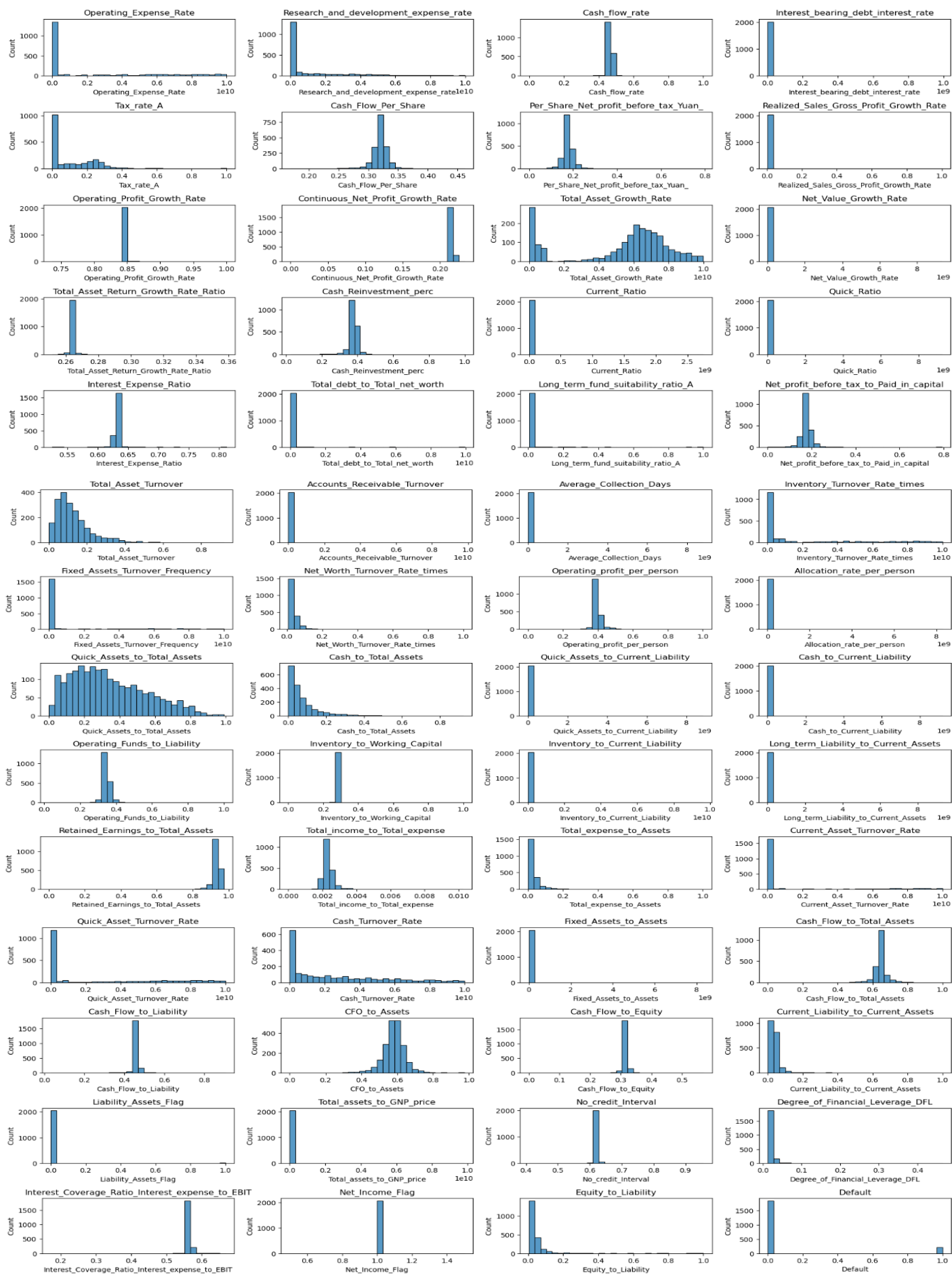


Figure 3

-Observations:

- There is no more normal distribution present in all the models, except CFO to assets shows bell shaped distribution that means it is a normal distribution.
- Most of the plots shows two or more peaks that means there are different groups are present in the dataset, and it indicate that a single model may not be adequate.

>Bivariate Analysis;

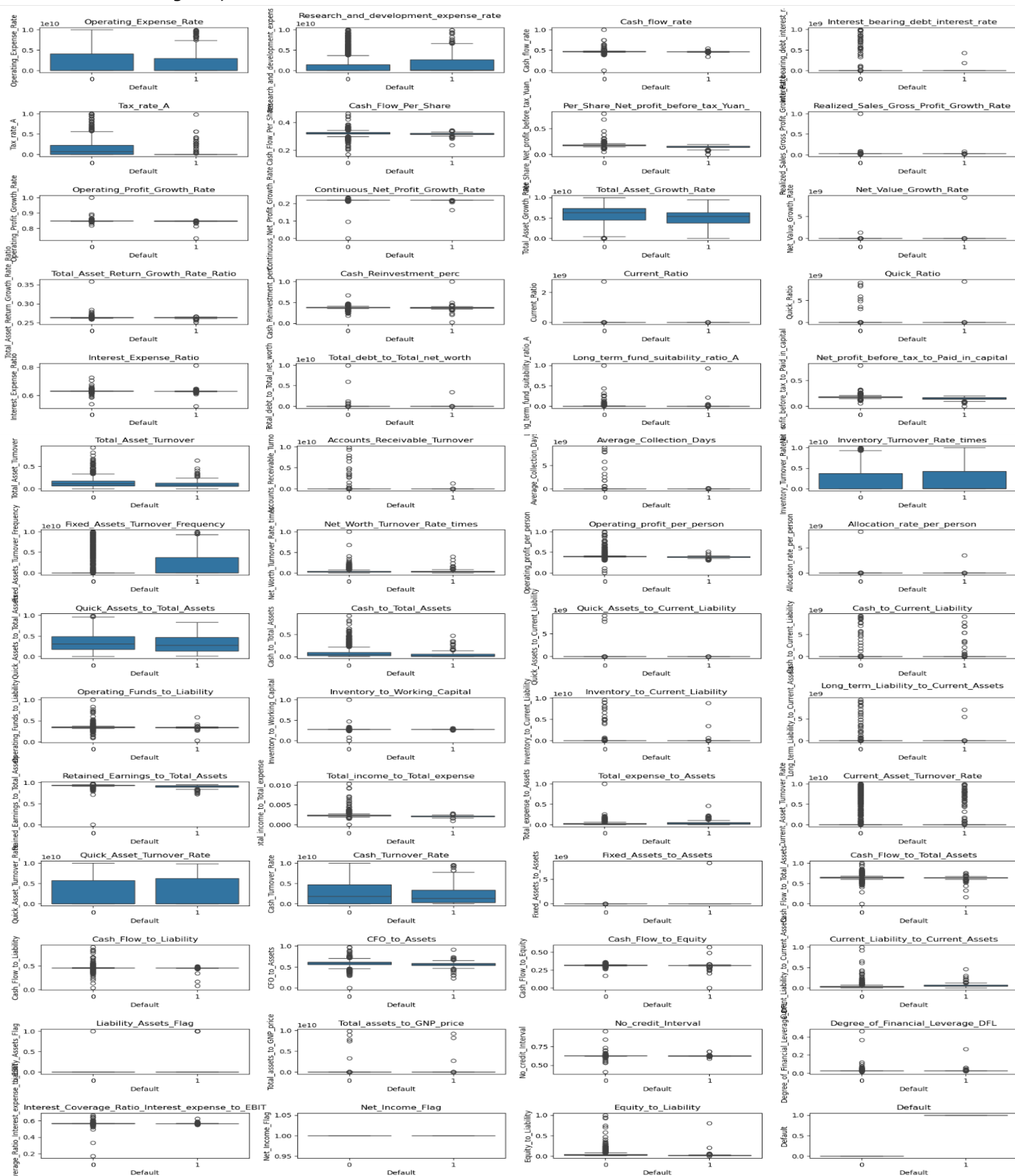


Figure 4

-Observations:

- All the box plots show so much outliers in the dataset.
- All the outliers indicate data are in issues and need further analysis.
- Some variables like "operating expense rate", "research and development expense rate", "total assets growth rate", "Inventory turnover rate times" and so more variables are strongly correlate with "default" outcome.
- The line inside each of the box represents the median of the numerical variables. Compare the medians for different categories in the "Default" column to understand how the central tendency varies.

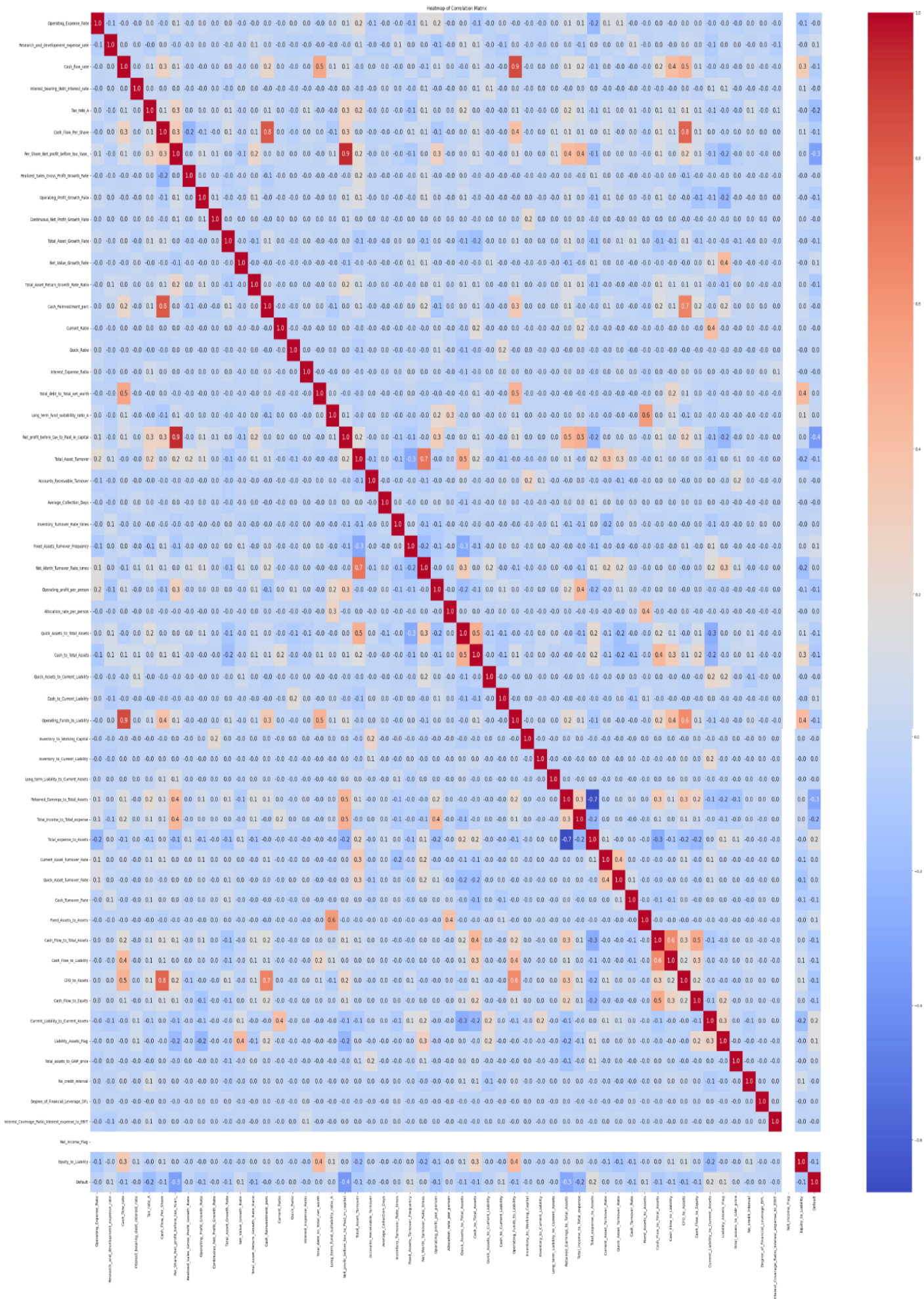


Figure 5

-Observations: From the above correlation plot we can say, it seems that the metrics are weekly correlated with each other.

2 - Data Pre-processing:

Dropping columns with few unique values.

```

Operating_Expense_Rate          1495
Research_and_development_expense_rate    629
Cash_flow_rate                  1888
Interest_bearing_debt_interest_rate      813
Tax_rate_A                      985
Cash_Flow_Per_Share              900
Per_Share_Net_profit_before_tax_Yuan_     876
Realized_Sales_Gross_Profit_Growth_Rate   1939
Operating_Profit_Growth_Rate             2015
Continuous_Net_Profit_Growth_Rate         2014
Total_Asset_Growth_Rate                 922
Net_Value_Growth_Rate                  1757
Total_Asset_Return_Growth_Rate_Ratio      1428
Cash_Reinvestment_perc                 1690
Current_Ratio                        1972
Quick_Ratio                          1970
Interest_Expense_Ratio               1716
Total_debt_to_Total_net_worth          1949
Long_term_fund_suitability_ratio_A       2014
Net_profit_before_tax_to_Paid_in_capital  1798
Total_Asset_Turnover                  283
Accounts_Receivable_Turnover           1109
Average_Collection_Days                1935
Inventory_Turnover_Rate_times           1151
Fixed_Assets_Turnover_Frequency         1079
Net_Worth_Turnover_Rate_times           529
Operating_profit_per_person             1484
Allocation_rate_per_person              2051
Quick_Assets_to_Total_Assets            2058
Cash_to_Total_Assets                   1962
Quick_Assets_to_Current_Liability        2058
Cash_to_Current_Liability               2056
Operating_Funds_to_Liability            2058
Inventory_to_Working_Capital             1931
Inventory_to_Current_Liability           1932
Long_term_Liability_to_Current_Assets    1398
Retained_Earnings_to_Total_Assets        2058
Total_income_to_Total_expense            2056
Total_expense_to_Assets                  2058
Current_Asset_Turnover_Rate              1973
Quick_Asset_Turnover_Rate                1743
Cash_Turnover_Rate                      1440
Fixed_Assets_to_Assets                  2054
Cash_Flow_to_Total_Assets                2058
Cash_Flow_to_Liability                   2058
CFO_to_Assets                           2058
Cash_Flow_to_Equity                     2058
Current_Liability_to_Current_Assets       2044
Liability_Assets_Flag                    2
Total_assets_to_GNP_price                 2058
No_credit_Interval                      2057
Degree_of_Financial_Leverage_DFL         1940
Interest_Coverage_Ratio_Interest_expense_to_EBIT  1945
Net_Income_Flag                          1
Equity_to_Liability                      2058
Default                                  2
dtype: int64

```

Table7- Unique Values of the numeric variables

We can drop the columns Net_Income_Flag and Liability_Assets_Flag as they have very few unique values.

```

Operating_Expense_Rate          1495
Research_and_development_expense_rate    629
Cash_flow_rate                  1888
Interest_bearing_debt_interest_rate      813
Tax_rate_A                      985
Cash_Flow_Per_Share             900
Per_Share_Net_profit_before_tax_Yuan_    876
Realized_Sales_Gross_Profit_Growth_Rate  1939
Operating_Profit_Growth_Rate          2015
Continuous_Net_Profit_Growth_Rate       2014
Total_Asset_Growth_Rate             922
Net_Value_Growth_Rate              1757
Total_Asset_Return_Growth_Rate_Ratio     1428
Cash_Reinvestment_perc              1690
Current_Ratio                     1972
Quick_Ratio                       1970
Interest_Expense_Ratio             1716
Total_debt_to_Total_net_worth          1949
Long_term_fund_suitability_ratio_A      2014
Net_profit_before_tax_to_Paid_in_capital 1798
Total_Asset_Turnover                283
Accounts_Receivable_Turnover          1109
Average_Collection_Days              1935
Inventory_Turnover_Rate_times          1151
Fixed_Assets_Turnover_Frequency        1079
Net_Worth_Turnover_Rate_times          529
Operating_profit_per_person           1484
Allocation_rate_per_person            2051
Quick_Assets_to_Total_Assets           2058
Cash_to_Total_Assets                 1962
Quick_Assets_to_Current_Liability      2058
Cash_to_Current_Liability             2056
Operating_Funds_to_Liability           2058
Inventory_to_Working_Capital            1931
Inventory_to_Current_Liability          1932
Long_term_Liability_to_Current_Assets   1398
Retained_Earnings_to_Total_Assets       2058
Total_income_to_Total_expense           2056
Total_expense_to_Assets                 2058
Current_Asset_Turnover_Rate             1973
Quick_Asset_Turnover_Rate              1743
Cash_Turnover_Rate                    1440
Fixed_Assets_to_Assets                 2054
Cash_Flow_to_Total_Assets              2058
Cash_Flow_to_Liability                 2058
CFO_to_Assets                         2058
Cash_Flow_to_Equity                    2058
Current_Liability_to_Current_Assets      2044
Total_assets_to_GNP_price               2058
No_credit_Interval                    2057
Degree_of_Financial_Leverage_DFL        1940
Interest_Coverage_Ratio_Interest_expense_to_EBIT 1945
Equity_to_Liability                   2058
Default                               2
dtype: int64

```

Table8- After dropping 2 columns the unique values of all the numeric variables

Outliers Check.

Number of outliers in each column:

	Column	No. of outliers
0	Operating_Expense_Rate	0
1	Research_and_development_expense_rate	264
2	Cash_flow_rate	206
3	Interest_bearing_debt_interest_rate	94
4	Tax_rate_A	42
5	Cash_Flow_Per_Share	146
6	Per_Share_Net_profit_before_tax_Yuan_	186
7	Realized_Sales_Gross_Profit_Growth_Rate	283
8	Operating_Profit_Growth_Rate	317
9	Continuous_Net_Profit_Growth_Rate	340
10	Total_Asset_Growth_Rate	0
11	Net_Value_Growth_Rate	304
12	Total_Asset_Return_Growth_Rate_Ratio	226
13	Cash_Reinvestment_perc	220
14	Current_Ratio	193
15	Quick_Ratio	190
16	Interest_Expense_Ratio	328
17	Total_debt_to_Total_net_worth	105
18	Long_term_fund_suitability_ratio_A	234
19	Net_profit_before_tax_to_Paid_in_capital	173
20	Total_Asset_Turnover	101
21	Accounts_Receivable_Turnover	281
22	Average_Collection_Days	77
23	Inventory_Turnover_Rate_times	29
24	Fixed_Assets_Turnover_Frequency	501
25	Net_Worth_Turnover_Rate_times	165
26	Operating_profit_per_person	357
27	Allocation_rate_per_person	200
28	Quick_Assets_to_Total_Assets	4
29	Cash_to_Total_Assets	163
30	Quick_Assets_to_Current_Liability	185
31	Cash_to_Current_Liability	253
32	Operating_Funds_to_Liability	219
33	Inventory_to_Working_Capital	247
34	Inventory_to_Current_Liability	129
35	Long_term_Liability_to_Current_Assets	213
36	Retained_Earnings_to_Total_Assets	208
37	Total_income_to_Total_expense	136
38	Total_expense_to_Assets	168
39	Current_Asset_Turnover_Rate	464
40	Quick_Asset_Turnover_Rate	0
41	Cash_Turnover_Rate	0
42	Fixed_Assets_to_Assets	10
43	Cash_Flow_to_Total_Assets	317
44	Cash_Flow_to_Liability	407
45	CFO_to_Assets	110
46	Cash_Flow_to_Equity	306
47	Current_Liability_to_Current_Assets	121
48	Total_assets_to_GNP_price	235
49	No_credit_Interval	396
50	Degree_of_Financial_Leverage_DFL	438
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	376
52	Equity_to_Liability	190
53	Default	220

Table9- Number of outliers in each column

-Observations:

- Key financial ratios like Operating Profit Growth Rate (317), Continuous Net Profit Growth Rate (340), and Interest Expense Ratio (328) have many outliers. This may indicate that some companies show unstable performance or extreme financial conditions.
- Metrics such as Operating Expense Rate, Total Asset Growth Rate, Quick Asset Turnover Rate, and Cash Turnover Rate show 0 outliers. This suggests these values are stable, which is good for reliable predictions in modeling.
- Metrics like Cash Flow to Liability (407) and Cash Flow to Total Assets (317) also have many outliers. This could point to some companies facing financial stress or showing unique strengths, warranting further investigation.
- The presence of many outliers in various metrics means they could affect analysis and modeling. It's important to consider methods like outlier removal or transformation to enhance model performance and clarity.

Data Preparation for Modeling.

- Separated target variable that is the "default" column from the rest of the dataset.
- Split the data into train and test.

Missing Values Detection and Treatment.

>X_train missing values;

Operating_Expense_Rate	0	
Research_and_development_expense_rate		0
Cash_flow_rate	0	
Interest_bearing_debt_interest_rate		0
Tax_rate_A	0	
Cash_Flow_Per_Share	126	
Per_Share_Net_profit_before_tax_Yuan_		0
Realized_Sales_Gross_Profit_Growth_Rate		0
Operating_Profit_Growth_Rate	0	
Continuous_Net_Profit_Growth_Rate		0
Total_Asset_Growth_Rate	0	
Net_Value_Growth_Rate	0	
Total_Asset_Return_Growth_Rate_Ratio		0
Cash_Reinvestment_perc	0	
Current_Ratio	0	
Quick_Ratio	0	
Interest_Expense_Ratio	0	
Total_debt_to_Total_net_worth	18	
Long_term_fund_suitability_ratio_A		0
Net_profit_before_tax_to_Paid_in_capital		0
Total_Asset_Turnover	0	
Accounts_Receivable_Turnover		0
Average_Collection_Days	0	
Inventory_Turnover_Rate_times		0
Fixed_Assets_Turnover_Frequency		0
Net_Worth_Turnover_Rate_times		0
Operating_profit_per_person	0	
Allocation_rate_per_person	0	
Quick_Assets_to_Total_Assets		0
Cash_to_Total_Assets	71	
Quick_Assets_to_Current_Liability		0
Cash_to_Current_Liability	0	
Operating_Funds_to_Liability	0	
Inventory_to_Working_Capital	0	
Inventory_to_Current_Liability	0	
Long_term_Liability_to_Current_Assets		0
Retained_Earnings_to_Total_Assets		0
Total_income_to_Total_expense		0
Total_expense_to_Assets	0	
Current_Asset_Turnover_Rate		0
Quick_Asset_Turnover_Rate		0
Cash_Turnover_Rate	0	
Fixed_Assets_to_Assets	0	
Cash_Flow_to_Total_Assets		0
Cash_Flow_to_Liability	0	
CFO_to_Assets	0	
Cash_Flow_to_Equity	0	
Current_Liability_to_Current_Assets		11
Total_assets_to_GNP_price		0
No_credit_Interval	0	
Degree_of_Financial_Leverage_DFL		0
Interest_Coverage_Ratio_Interest_expense_to_EBIT		0
Equity_to_Liability	0	

dtype: int64

Table10- X_train missing value detection

-Observations:

- Most metrics show 0 outliers, indicating that many financial ratios are stable across the dataset. This suggests that these metrics are consistent and may provide reliable data for analysis.
- The metrics Cash Flow Per Share (126) and Cash to Total Assets (71) have some outliers, while Total debt to Total net worth (18) and Current Liability to Current Assets (11) also have a few. These outliers may reflect variability in financial performance and should be examined further for potential insights.

>X_test missing values;

Operating_Expense_Rate	0
Research_and_development_expense_rate	0
Cash_flow_rate	0
Interest_bearing_debt_interest_rate	0
Tax_rate_A	0
Cash_Flow_Per_Share	41
Per_Share_Net_profit_before_tax_Yuan_	0
Realized_Sales_Gross_Profit_Growth_Rate	0
Operating_Profit_Growth_Rate	0
Continuous_Net_Profit_Growth_Rate	0
Total_Asset_Growth_Rate	0
Net_Value_Growth_Rate	0
Total_Asset_Return_Growth_Rate_Ratio	0
Cash_Reinvestment_perc	0
Current_Ratio	0
Quick_Ratio	0
Interest_Expense_Ratio	0
Total_debt_to_Total_net_worth	3
Long_term_fund_suitability_ratio_A	0
Net_profit_before_tax_to_Paid_in_capital	0
Total_Asset_Turnover	0
Accounts_Receivable_Turnover	0
Average_Collection_Days	0
Inventory_Turnover_Rate_times	0
Fixed_Assets_Turnover_Frequency	0
Net_Worth_Turnover_Rate_times	0
Operating_profit_per_person	0
Allocation_rate_per_person	0
Quick_Assets_to_Total_Assets	0
Cash_to_Total_Assets	25
Quick_Assets_to_Current_Liability	0
Cash_to_Current_Liability	0
Operating_Funds_to_Liability	0
Inventory_to_Working_Capital	0
Inventory_to_Current_Liability	0
Long_term_Liability_to_Current_Assets	0
Retained_Earnings_to_Total_Assets	0
Total_income_to_Total_expense	0
Total_expense_to_Assets	0
Current_Asset_Turnover_Rate	0
Quick_Asset_Turnover_Rate	0
Cash_Turnover_Rate	0
Fixed_Assets_to_Assets	0
Cash_Flow_to_Total_Assets	0
Cash_Flow_to_Liability	0
CFO_to_Assets	0
Cash_Flow_to_Equity	0
Current_Liability_to_Current_Assets	3
Total_assets_to_GNP_price	0
No_credit_Interval	0
Degree_of_Financial_Leverage_DFL	0
Interest_Coverage_Ratio_Interest_expense_to_EBIT	0
Equity_to_Liability	0

dtype: int64

Table11- X_test missing value detection

-Observations:

- Most financial metrics show 0 outliers, indicating that the majority of these values are stable and consistent. This consistency is beneficial for reliable analysis and modeling.
- The metrics Cash Flow Per Share (41) and Cash to Total Assets (25) have a moderate number of outliers, while Total debt to Total net worth (3) and Current Liability to Current Assets (3) have very few. These outliers might indicate some variability in financial performance that could warrant further investigation.
- Replaced the missing values using KNN imputer.

Scaling the Data.

	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_I
0	-0.633296	-0.396806	-0.132455	-0.128462	-0.754347	0.088170	
1	-0.633296	-0.561672	-0.934352	-0.128462	-0.754347	-1.224514	
2	-0.633296	0.361946	-0.290335	-0.128462	0.061964	-0.409659	
3	-0.633296	-0.561672	-0.179548	-0.128462	-0.754347	-0.077773	
4	-0.633296	-0.561672	-0.123892	-0.128462	-0.754347	-0.168422	

Table12- Head of the X_train_scaled data

	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_I
0	-0.633296	1.539557	0.118477	-0.128462	-0.754347	0.053496	
1	2.016795	0.135659	0.441752	-0.128462	-0.754347	0.164950	
2	-0.633296	0.177222	-0.141279	-0.128462	0.085538	0.112938	
3	-0.633296	2.144527	-0.666470	-0.128462	-0.754347	-0.986745	
4	0.763746	0.301910	-0.050325	-0.128462	0.719997	0.578569	

Table13- Head of the X_test_scaled dada

3 - Model Building:

Model Evaluation Criterion.

>Metric of Choice;

Logistic Regression.

	const	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_I
0	1.0	-0.633296	-0.396806	-0.132455	-0.128462	-0.754347	0.088170	
1	1.0	-0.633296	-0.561672	-0.934352	-0.128462	-0.754347	-1.224514	
2	1.0	-0.633296	0.361946	-0.290335	-0.128462	0.061964	-0.409659	
3	1.0	-0.633296	-0.561672	-0.179548	-0.128462	-0.754347	-0.077773	
4	1.0	-0.633296	-0.561672	-0.123892	-0.128462	-0.754347	-0.168422	

5 rows × 54 columns

Table14- Head of the X_train_with_intercept

Warning: Maximum number of iterations has been exceeded.
Current function value: 0.193946
Iterations: 35

Logit Regression Results

```

=====
Dep. Variable:          Default      No. Observations:      1543
Model:                  Logit        Df Residuals:          1489
Method:                 MLE          Df Model:              53
Date:                  Thu, 17 Oct 2024      Pseudo R-squ.:        0.4297
Time:                  11:35:47      Log-Likelihood:        -299.26
Converged:              False        LL-Null:              -524.71
Covariance Type:       nonrobust      LLR p-value:          1.764e-64
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	-7.4685	2410.789	-0.003	0.998	-4732.528	4717.591
Operating_Expense_Rate	0.2077	0.121	1.713	0.087	-0.030	0.445
Research_and_development_expense_rate	0.3556	0.104	3.433	0.001	0.153	0.559
Cash_flow_rate	-0.1837	1.016	-0.181	0.857	-2.175	1.808
Interest_bearing_debt_interest_rate	0.1755	0.151	1.163	0.245	-0.120	0.471
Tax_rate_A	-0.2580	0.174	-1.481	0.139	-0.599	0.083
Cash_Flow_Per_Share	-0.3533	0.281	-1.260	0.208	-0.903	0.196
Per_Share_Net_profit_before_tax_Yuan_	0.2518	1.276	0.197	0.844	-2.249	2.752
Realized_Sales_Gross_Profit_Growth_Rate	0.1012	0.118	0.859	0.390	-0.130	0.332
Operating_Profit_Growth_Rate	-0.1546	0.267	-0.579	0.563	-0.678	0.369
Continuous_Net_Profit_Growth_Rate	0.1736	0.132	1.317	0.188	-0.085	0.432
Total_Asset_Growth_Rate	-0.0640	0.131	-0.487	0.626	-0.321	0.193
Net_Value_Growth_Rate	0.5177	3097.843	0.000	1.000	-6071.142	6072.178
Total_Asset_Return_Growth_Rate_Ratio	-0.3299	0.361	-0.915	0.360	-1.037	0.377
Cash_Reinvestment_perc	0.1700	0.346	0.491	0.624	-0.509	0.849
Current_Ratio	-1.6114	0.925	-1.742	0.081	-3.424	0.201
Quick_Ratio	-2.7355	2.57e+04	-0.000	1.000	-5.05e+04	5.05e+04
Interest_Expense_Ratio	0.0197	0.065	0.303	0.762	-0.107	0.147
Total_debt_to_Total_net_worth	1.9035	0.623	3.058	0.002	0.683	3.124
Long_term_fund_suitability_ratio_A	0.1675	0.223	0.751	0.452	-0.269	0.604
Net_profit_before_tax_to_Paid_in_capital	-1.0834	1.179	-0.919	0.358	-3.394	1.227
Total_Asset_Turnover	-0.2122	0.319	-0.666	0.506	-0.837	0.413
Accounts_Receivable_Turnover	-1.0019	0.642	-1.560	0.119	-2.261	0.257
Average_Collection_Days	-15.1938	2.49e+04	-0.001	1.000	-4.89e+04	4.88e+04
Inventory_Turnover_Rate_times	-0.0490	0.117	-0.420	0.675	-0.278	0.180
Fixed_Assets_Turnover_Frequency	0.1775	0.106	1.678	0.093	-0.030	0.385
Net_Worth_Turnover_Rate_times	-0.2559	0.211	-1.212	0.225	-0.670	0.158
Operating_profit_per_person	0.0505	0.195	0.259	0.796	-0.331	0.432
Allocation_rate_per_person	-80.4894	153.635	-0.524	0.600	-381.609	220.630
Quick_Assets_to_Total_Assets	0.1935	0.189	1.024	0.306	-0.177	0.564
Cash_to_Total_Assets	-0.3059	0.222	-1.380	0.168	-0.740	0.129
Quick_Assets_to_Current_Liability	-0.5860	1.49e+04	-3.92e-05	1.000	-2.93e+04	2.93e+04
Cash_to_Current_Liability	0.0684	0.076	0.905	0.365	-0.080	0.217
Operating_Funds_to_Liability	1.2409	0.783	1.584	0.113	-0.294	2.776
Inventory_to_Working_Capital	-0.1714	0.158	-1.088	0.276	-0.480	0.137
Inventory_to_Current_Liability	0.1022	0.117	0.870	0.384	-0.128	0.332
Long_term_Liability_to_Current_Assets	-0.0208	0.107	-0.195	0.846	-0.230	0.188
Retained_Earnings_to_Total_Assets	-0.2111	0.207	-1.019	0.308	-0.617	0.195
Total_income_to_Total_expense	-1.4219	0.437	-3.252	0.001	-2.279	-0.565
Total_expense_to_Assets	0.0849	0.253	0.335	0.738	-0.412	0.582
Current_Asset_Turnover_Rate	-0.0962	0.129	-0.746	0.456	-0.349	0.157
Quick_Asset_Turnover_Rate	0.0640	0.128	0.499	0.618	-0.188	0.316
Cash_Turnover_Rate	-0.4286	0.130	-3.307	0.001	-0.683	-0.175
Fixed_Assets_to_Assets	31.5360	195.727	0.161	0.872	-352.083	415.155
Cash_Flow_to_Total_Assets	0.9901	0.270	3.668	0.000	0.461	1.519
Cash_Flow_to_Liability	-2.7554	0.607	-4.542	0.000	-3.945	-1.566
CFO_to_Assets	-0.3143	0.467	-0.673	0.501	-1.230	0.602
Cash_Flow_to_Equity	-0.0344	0.085	-0.404	0.686	-0.201	0.132
Current_Liability_to_Current_Assets	-0.0863	0.121	-0.714	0.476	-0.323	0.151
Total_assets_to_GNP_price	-0.0290	0.076	-0.384	0.701	-0.177	0.119
No_credit_Interval	0.1051	0.079	1.326	0.185	-0.050	0.260
Degree_of_Financial_Leverage_DFL	0.0729	0.056	1.303	0.193	-0.037	0.183
Interest_Coverage_Ratio_Interest_expense_to_EBIT	0.0677	0.087	0.778	0.437	-0.103	0.238
Equity_to_Liability	-3.0217	0.709	-4.260	0.000	-4.412	-1.632

>Logistic Regression Model - Training Performance;

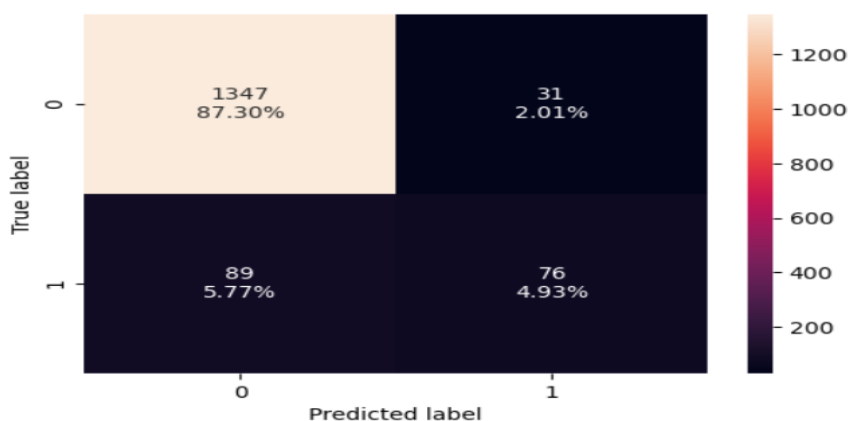


Figure 6- Logistic Regression Model(Training Performance)

	Accuracy	Recall	Precision	F1
0	0.922229	0.460606	0.71028	0.558824

-Observations:

- The model is 92.2% accurate, meaning it generally makes correct predictions. However, its recall is only 46.1%, which means it's missing many positive cases. This is important because it suggests the model struggles to identify instances that matter, potentially leading to significant oversights.
- The model has a precision of 71.0%, indicating that when it predicts a positive case, it's correct 71% of the time. However, the F1 score of 55.9% shows that there's room for improvement in balancing both recall and precision, especially in finding those important positive cases.

>Logistic Regression Model - Test Performance;

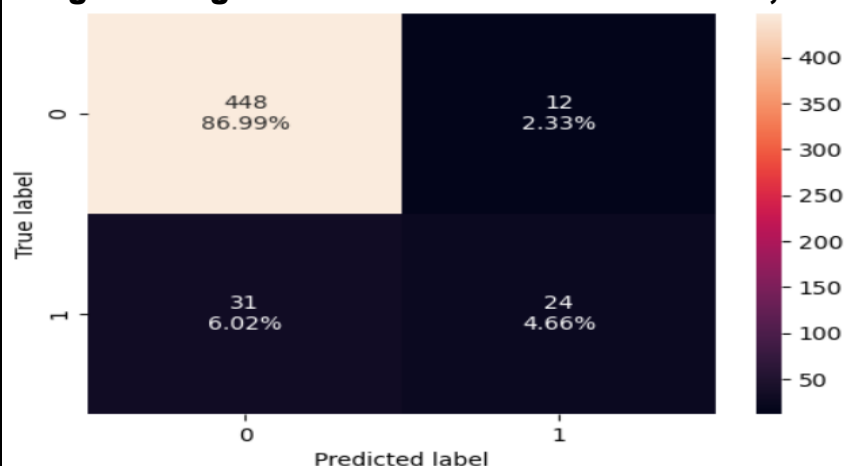


Figure 7- Logistic Regression Mode(Test Performance)

	Accuracy	Recall	Precision	F1
0	0.916505	0.436364	0.666667	0.527473

-Observations:

- The model is 91.7% accurate, meaning it usually makes correct predictions. However, its recall is only 43.6%, indicating it often misses important positive cases. This can be a problem if identifying those positives is essential.
- The precision is 66.7%, so when the model predicts a positive case, it's right about two-thirds of the time. The F1 score of 52.7% shows there's a significant need for improvement in the model's ability to find and correctly predict positive instances.

Random Forest.

>Random Forest Model - Training Performance;\

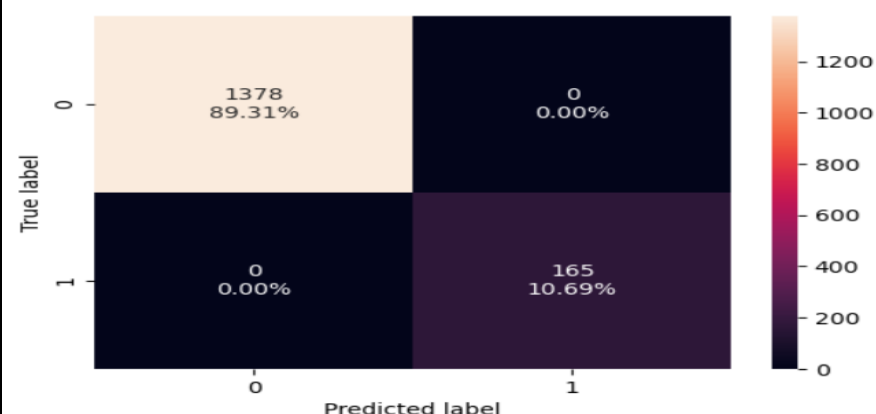


Figure 8- Random Forest Model(Training Performance)

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

-Observations:

- The model has an accuracy of 100%, meaning it predicts every case correctly. It also has 100% recall and precision, indicating it identifies all positive cases without any mistakes.
- The F1 score of 1.0 shows that the model is perfectly balanced in finding and correctly predicting positive cases.

>Random Forest Model - Test Performance;

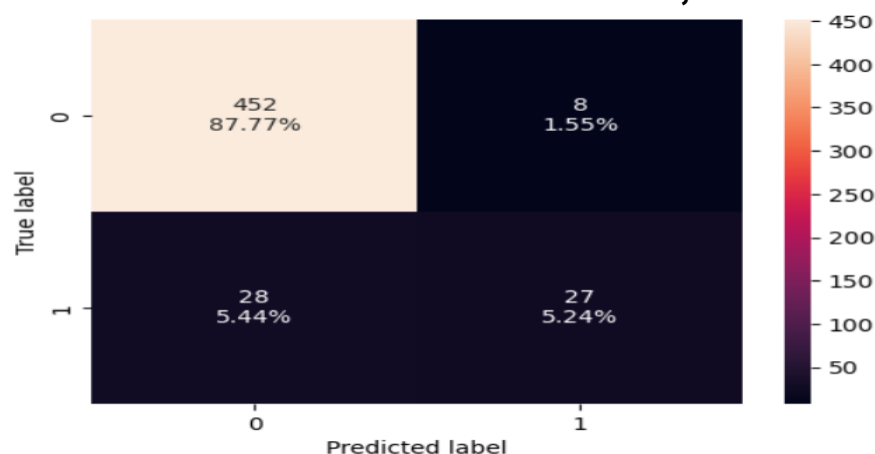


Figure9- Random Forest Model(Test Performance)

	Accuracy	Recall	Precision	F1
0	0.930097	0.490909	0.771429	0.6

-Observations:

- The model has an accuracy of 93.0%, meaning it gets most predictions right. However, with a recall of 49.1%, it misses about half of the positive cases, which is concerning if finding those positives is important.
- The precision of 77.1% indicates that when the model predicts a positive case, it's correct 77% of the time. While this is reasonable, the low recall suggests it isn't capturing all the positive instances effectively.
- The F1 score of 0.6 shows there's a need for improvement. It reflects the balance between precision and recall, indicating that the model could perform better in identifying positive cases.

4 - Model Performance Improvement:

Model Performance Improvement - Logistic Regression.

>Variance Inflation Factor (VIF) calculation;

Variance Inflation Factors:

	Variable	VIF
0	Operating_Expense_Rate	1.259611
1	Research_and_development_expense_rate	1.099306
2	Cash_flow_rate	12.259184
3	Interest_bearing_debt_interest_rate	1.032597
4	Tax_rate_A	1.248346
5	Cash_Flow_Per_Share	4.564430
6	Per_Share_Net_profit_before_tax_Yuan_	8.756612
7	Realized_Sales_Gross_Profit_Growth_Rate	1.058517
8	Operating_Profit_Growth_Rate	1.152581
9	Continuous_Net_Profit_Growth_Rate	1.467948
10	Total_Asset_Growth_Rate	1.174794
11	Net_Value_Growth_Rate	1.044367
12	Total_Asset_Return_Growth_Rate_Ratio	1.134845
13	Cash_Reinvestment_perc	7.340538
14	Current_Ratio	4.945713
15	Quick_Ratio	1.063685
16	Interest_Expense_Ratio	1.033551
17	Total_debt_to_Total_net_worth	3.776391
18	Long_term_fund_suitability_ratio_A	1.839945
19	Net_profit_before_tax_to_Paid_in_capital	8.637685
20	Total_Asset_Turnover	5.467530
21	Accounts_Receivable_Turnover	1.064519
22	Average_Collection_Days	1.060724
23	Inventory_Turnover_Rate_times	1.100171
24	Fixed_Assets_Turnover_Frequency	1.223623
25	Net_Worth_Turnover_Rate_times	3.945259
26	Operating_profit_per_person	1.568575
27	Allocation_rate_per_person	1.198618
28	Quick_Assets_to_Total_Assets	2.397607
29	Cash_to_Total_Assets	2.183010
30	Quick_Assets_to_Current_Liability	1.009579
31	Cash_to_Current_Liability	1.079209
32	Operating_Funds_to_Liability	12.536226
33	Inventory_to_Working_Capital	1.459350
34	Inventory_to_Current_Liability	1.124100
35	Long_term_Liability_to_Current_Assets	1.102010
36	Retained_Earnings_to_Total_Assets	3.365775
37	Total_income_to_Total_expense	1.676735
38	Total_expense_to_Assets	3.366230
39	Current_Asset_Turnover_Rate	1.416203
40	Quick_Asset_Turnover_Rate	1.377544
41	Cash_Turnover_Rate	1.107230
42	Fixed_Assets_to_Assets	1.815190
43	Cash_Flow_to_Total_Assets	3.309496
44	Cash_Flow_to_Liability	2.813638
45	CFO_to_Assets	10.987676
46	Cash_Flow_to_Equity	1.425404
47	Current_Liability_to_Current_Assets	1.464422
48	Total_assets_to_GNP_price	1.041114
49	No_credit_Interval	1.032531
50	Degree_of_Financial_Leverage_DFL	1.015012
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	1.018535
52	Equity_to_Liability	4.779776

Table15- VIF calculation

-Some High VIF columns

['Cash_flow_rate',
'Per_Share_Net_profit_before_tax_Yuan_',
'Cash_Reinvestment_perc',
'Net_profit_before_tax_to_Paid_in_capital',
'Total_Asset_Turnover',
'Operating_Funds_to_Liability',
'CFO_to_Assets']

-Observations:

- Cash_flow_rate is 12.26, Operating_Funds_to_Liability is 12.54, and CFO_to_Assets is 10.99 have high VIFs. A VIF over 10 indicates that all these variables are likely correlated with others, and that can make regression results unreliable.
- Variables like Cash_Flow_Per_Share is 4.56, Per_Share_Net_profit_before_tax_Yuan_ is 8.76, and Net_profit_before_tax_to_Paid_in_capital is 8.64 show moderate correlation. While not as concerning as the high VIFs, but they also need attention for further.
- We can see many variables, including Operating_Expense_Rate is 1.26 and Research_and_development_expense_rate is 1.10, have low VIFs. This means they are fairly independent and should provide reliable insights for the model.

-After dropping high VIF values columns the X_train_scaled data contain 1543 rows and 46 columns.

-After dropping high VIF values columns the X_test_scaled data contain 515 rows and 46 columns.

-Now retraining logistic regression model with new data

Current function value: 0.201099
Iterations: 100
Function evaluations: 101
Gradient evaluations: 101

Logit Regression Results

Dep. Variable:	Default	No. Observations:	1543
Model:	Logit	Df Residuals:	1496
Method:	MLE	Df Model:	46
Date:	Thu, 17 Oct 2024	Pseudo R-squ.:	0.4086
Time:	11:36:44	Log-Likelihood:	-310.30
converged:	False	LL-Null:	-524.71
Covariance Type:	nonrobust	LLR p-value:	1.462e-63

	coef	std err	z	P> z	[0.025	0.975]
const	-4.4843	0.730	-6.144	0.000	-5.915	-3.054
Operating_Expense_Rate	0.1938	0.117	1.658	0.097	-0.035	0.423
Research_and_development_expense_rate	0.3735	0.099	3.759	0.000	0.179	0.568
Interest_bearing_debt_interest_rate	0.1971	0.152	1.301	0.193	-0.100	0.494
Tax_rate_A	-0.3721	0.180	-2.066	0.039	-0.725	-0.019
Cash_Flow_Per_Share	-0.1821	0.139	-1.312	0.189	-0.454	0.090
Realized_Sales_Gross_Profit_Growth_Rate	0.1174	0.116	1.012	0.312	-0.110	0.345
Operating_Profit_Growth_Rate	-0.2278	0.300	-0.759	0.448	-0.816	0.360
Continuous_Net_Profit_Growth_Rate	0.1505	0.123	1.227	0.220	-0.090	0.391
Total_Asset_Growth_Rate	-0.0667	0.126	-0.531	0.595	-0.313	0.179
Net_Value_Growth_Rate	0.1937	3.545	0.055	0.956	-6.754	7.142
Total_Asset_Return_Growth_Rate_Ratio	-0.3428	0.373	-2.063	0.039	-1.502	-0.039
Current_Ratio	-1.9304	0.643	-3.000	0.003	-3.192	-0.669
Quick_Ratio	-0.7513	7.542	-0.100	0.921	-15.533	14.030
Interest_Expense_Ratio	0.0258	0.065	0.396	0.692	-0.102	0.154
Total_debt_to_Total_net_worth	2.8590	0.569	5.021	0.000	1.743	3.975
Long_term_fund_suitability_ratio_A	-0.2377	0.253	-0.938	0.348	-0.734	0.259
Accounts_Receivable_Turnover	-1.0112	0.619	-1.634	0.102	-2.224	0.202
Average_Collection_Days	-0.3428	1.827	-0.188	0.851	-3.923	3.237
Inventory_Turnover_Rate_times	-0.0581	0.114	-0.511	0.609	-0.281	0.165
Fixed_Assets_Turnover_Frequency	0.1469	0.104	1.417	0.157	-0.056	0.350
Net_Worth_Turnover_Rate_times	-0.1894	0.129	-1.472	0.141	-0.442	0.063
Operating_profit_per_person	0.0322	0.187	0.172	0.864	-0.335	0.400
Allocation_rate_per_person	-0.0413	1.387	-0.030	0.976	-2.759	2.677
Quick_Assets_to_Total_Assets	0.0429	0.161	0.266	0.790	-0.273	0.359
Cash_to_Total_Assets	-0.3529	0.212	-1.666	0.096	-0.768	0.062
Quick_Assets_to_Current_Liability	-0.1175	1.661	-0.071	0.944	-3.372	3.137
Cash_to_Current_Liability	0.0739	0.075	0.992	0.321	-0.072	0.220
Inventory_to_Working_Capital	-0.1518	0.143	-1.058	0.290	-0.433	0.129
Inventory_to_Current_Liability	0.0899	0.124	0.724	0.469	-0.153	0.333
Long_term_Liability_to_Current_Assets	-0.0475	0.108	-0.439	0.661	-0.259	0.165
Retained_Earnings_to_Total_Assets	-0.2175	0.179	-1.215	0.224	-0.568	0.133
Total_income_to_Total_expense	-2.0469	0.354	-5.783	0.000	-2.741	-1.353
Total_expense_to_Assets	0.1727	0.206	0.837	0.403	-0.232	0.577
Current_Asset_Turnover_Rate	-0.1299	0.120	-1.086	0.277	-0.364	0.104
Quick_Asset_Turnover_Rate	0.0295	0.120	0.247	0.805	-0.205	0.264
Cash_Turnover_Rate	-0.3696	0.123	-3.015	0.003	-0.610	-0.129
Fixed_Assets_to_Assets	0.5126	17.419	0.029	0.977	-33.628	34.653
Cash_Flow_to_Total_Assets	0.9891	0.232	4.269	0.000	0.535	1.443
Cash_Flow_to_Liability	-2.2925	0.452	-5.077	0.000	-3.177	-1.408
Cash_Flow_to_Equity	0.0059	0.073	0.082	0.935	-0.136	0.148
Current_Liability_to_Current_Assets	-0.0785	0.115	-0.685	0.494	-0.303	0.146
Total_assets_to_GNP_price	-0.0347	0.075	-0.463	0.643	-0.182	0.112
No_credit_Interval	0.1180	0.080	1.476	0.140	-0.039	0.275
Degree_of_Financial_Leverage_DFL	0.0772	0.055	1.403	0.161	-0.031	0.185
Interest_Coverage_Ratio_Interest_expense_to_EBIT	0.0597	0.082	0.727	0.467	-0.101	0.221
Equity_to_Liability	-2.9001	0.554	-5.236	0.000	-3.986	-1.814

-The Optimal Threshold Value is 0.084.

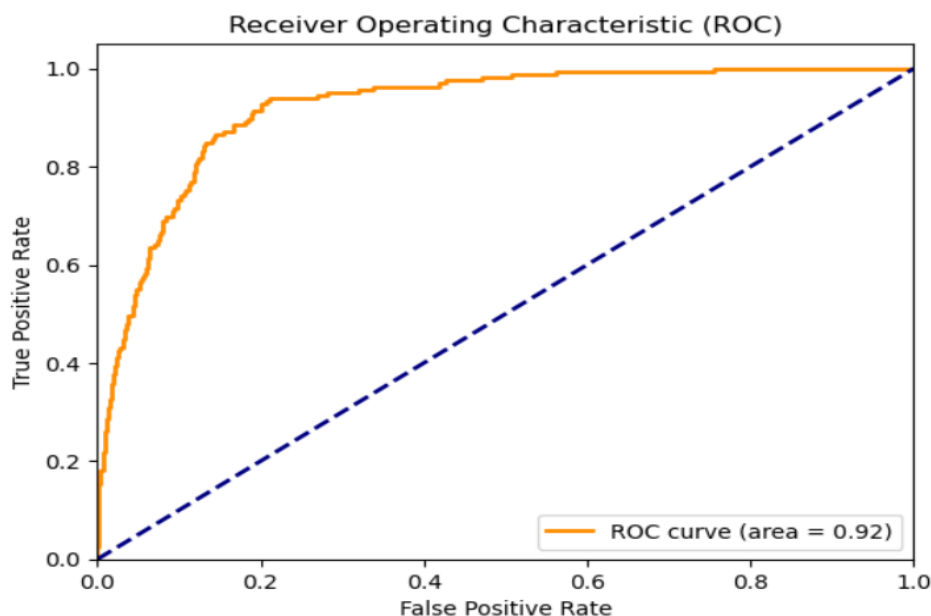


Figure 10- ROC score

-Observations: The model performance is excellent because the roc score is 0.92.

>Logistic Regression Performance - Training Set;

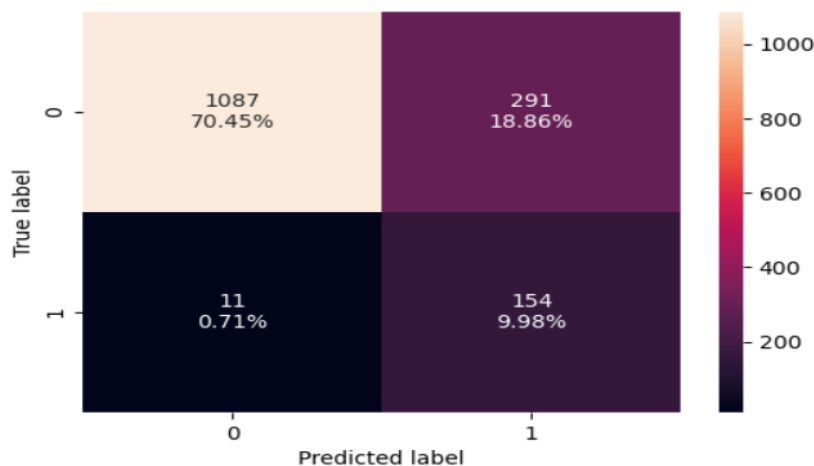


Figure 11- Logistic Regression Performance - Training Set

	Accuracy	Recall	Precision	F1
0	0.804277	0.933333	0.346067	0.504918

-Observations:

- The model correctly predicted outcomes for about 80.43% of cases. While this is a good level of accuracy, it's important to look at how well it performs for each class.
- The model has a high recall of 93.33%, that mean it effectively identifies most actual positive cases. This is especially important in situations where missing a positive case could have serious consequences, like in fraud detection.
- Precision is lower at 34.61%. It means that many of the positive predictions made by the model are incorrect. As a result, actions based on these predictions might be very costly.
- The F1 score, which balances precision and recall, is 50.49%. Which indicates that there is significant room for improvement in making the model more reliable.

>Logistic Regression Performance - Test Set;

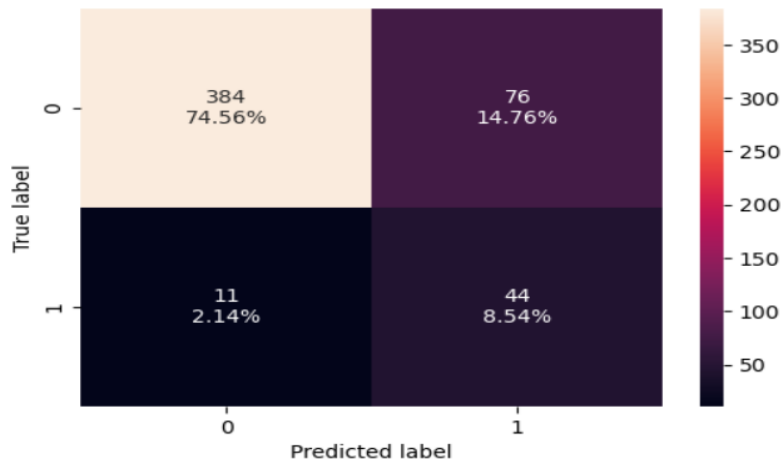


Figure 12- Logistic Regression Performance - Test Set

	Accuracy	Recall	Precision	F1
0	0.831068	0.8	0.366667	0.502857

-Observations:

- The model correctly predicts about 83 out of 100 cases. That means its shows that the model is generally effective, but we need to look at how well it performs for different types of outcomes.
- The model identifies 80% of the actual positive cases. Which is good, especially in important situations like fraud detection.
- Precision is low at 36.67%. This means that only about one-third of the cases the model predicts as positive are actually correct.
- The F1 score is 50.29%. This score shows there's a balance between recall and precision, but it also highlights that there's significant room for improvement, especially in precision.

Model Performance Improvement - Random Forest.

Parameters used in the Random Forest Classifier:

```
bootstrap: True
ccp_alpha: 0.0
class_weight: balanced
criterion: gini
max_depth: 5
max_features: sqrt
max_leaf_nodes: None
max_samples: None
min_impurity_decrease: 0.0
min_samples_leaf: 7
min_samples_split: 2
min_weight_fraction_leaf: 0.0
monotonic_cst: None
n_estimators: 200
n_jobs: None
oob_score: False
random_state: 42
verbose: 0
warm_start: False
```

>Random Forest Performance - Training Set;

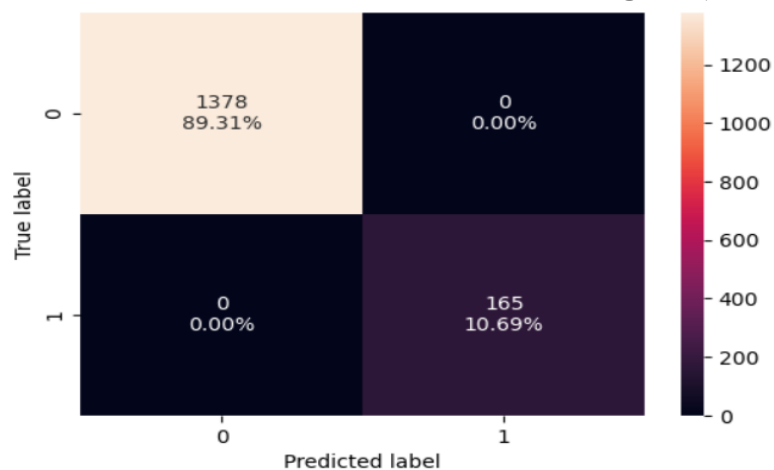


Figure 13- Random Forest Performance - Training Set

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

-Observations: here the Accuracy, Recall, Precision and F1 score is 100%

- The model correctly predicted all outcomes, showing perfect accuracy. This means it is highly reliable.
- The model identified every actual positive case. This is vital in important areas like healthcare or fraud detection, where missing a positive case can have serious consequences.
- Every prediction made as positive was correct, meaning there are no false positives. This is great for efficient decision-making and resource use.
- The F1 score is also perfect, indicating that the model performs exceptionally well across all key measures.

>Random Forest Performance - Test Set;

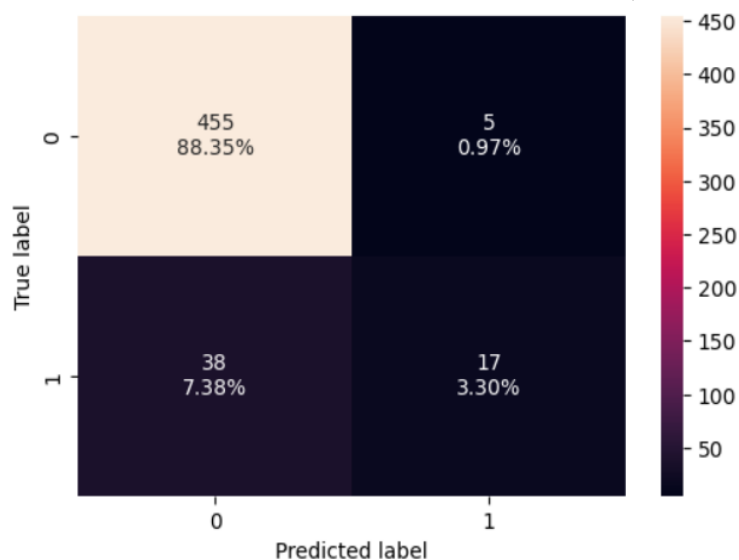


Figure 14- Random Forest Performance - Test Set

	Accuracy	Recall	Precision	F1
0	0.916505	0.309091	0.772727	0.441558

-Observations:

- The model correctly predicted about 92 out of 100 cases. This is good overall, but we need to look deeper into how well it performs for different outcomes.
- The recall rate is low at 30.91%, that means the model only identifies about 31% of actual positive cases. In important situations like fraud detection.
- The precision score is higher at 77.27%. It means that when the model predicts a positive case, it is correct about 77% of the time. However, this indicates a trade-off, as the model may be missing many positives.
- The F1 score is 44.16%, basically it balances precision and recall. It shows that while the model is fairly accurate when it predicts positives, the low recall is a significant concern.

5 - Model Performance Comparison and Final Model Selection:

>Training Performance Comparison;

Training performance comparison:

	Logistic Regression	Tuned Logistic Regression	Random Forest	Tuned Random Forest
Accuracy	0.922229	0.804277	1.0	1.0
Recall	0.460606	0.933333	1.0	1.0
Precision	0.710280	0.346067	1.0	1.0
F1	0.558824	0.504918	1.0	1.0

Table16- Training Performance Comparison

-Observations:

- Random Forest and Tuned Random Forest both achieved perfect accuracy (100%), meaning they predicted every outcome correctly.
- Logistic Regression performed well with an accuracy of 92.22%. However, the Tuned Logistic Regression dropped to 80.43%, indicating a decrease in overall prediction effectiveness.
- Tuned Logistic Regression improved recall to 93.33%, successfully identifying a high percentage of actual positive cases.
- In contrast, Logistic Regression only captured 46.06% of positives, which is concerning.
- Both Random Forest models excelled with perfect recall (100%), meaning they identified all actual positive cases.
- Logistic Regression had a good precision of 71.03%, meaning most of its positive predictions were correct.
- However, Tuned Logistic Regression had a much lower precision of 34.61%, suggesting that many of its positive predictions were wrong.
- Both Random Forest models achieved perfect precision (100%), which is excellent for making reliable decisions.
- The F1 score balances precision and recall. Logistic Regression scored 55.88%, while Tuned Logistic Regression scored 50.49%.
- The Random Forest models scored perfectly (100%), indicating they provide a strong balance and reliability in predictions.

>Testing Performance Comparison;

Testing performance comparison:

	Logistic Regression	Tuned Logistic Regression	Random Forest	Tuned Random Forest
Accuracy	0.916505	0.831068	0.930097	0.916505
Recall	0.436364	0.800000	0.490909	0.309091
Precision	0.666667	0.366667	0.771429	0.772727
F1	0.527473	0.502857	0.600000	0.441558

Table17- Testing Performance Comparison

-Observations:

- Random Forest score is 93.01% and Logistic Regression score is 91.65% which showed high accuracy, indicating strong overall predictive performance. Tuned Logistic Regression score 83.11% which had a lower accuracy, while Tuned Random Forest score is 91.65% that means it is performed similarly to Logistic Regression.
- Tuned Logistic Regression score is 80.00% which is excelled in recall, effectively identifying a higher percentage of actual positive cases. Random Forest score is 49.09% and Logistic Regression score is 43.64% which had moderate recall, while Tuned Random Forest score 30.91% lagged significantly, suggesting a risk of missing important positive instances.
- Random Forest score is 77.14% and Tuned Random Forest score is 77.27% which maintained high precision, that indicate the most of their positive predictions were correct. In contrast, Logistic Regression score is 66.67% that is performed better than Tuned Logistic Regression score 36.67%, which indicates many incorrect positive predictions.
- The F1 scores reflect a balance between precision and recall. Random Forest score is 60.00% that is performed well, while Logistic Regression score is 52.75% also showed decent performance. Tuned Logistic Regression score is a similar score that is 50.29%, but Tuned Random Forest score is 44.16% which is indicate that an imbalance in performance.

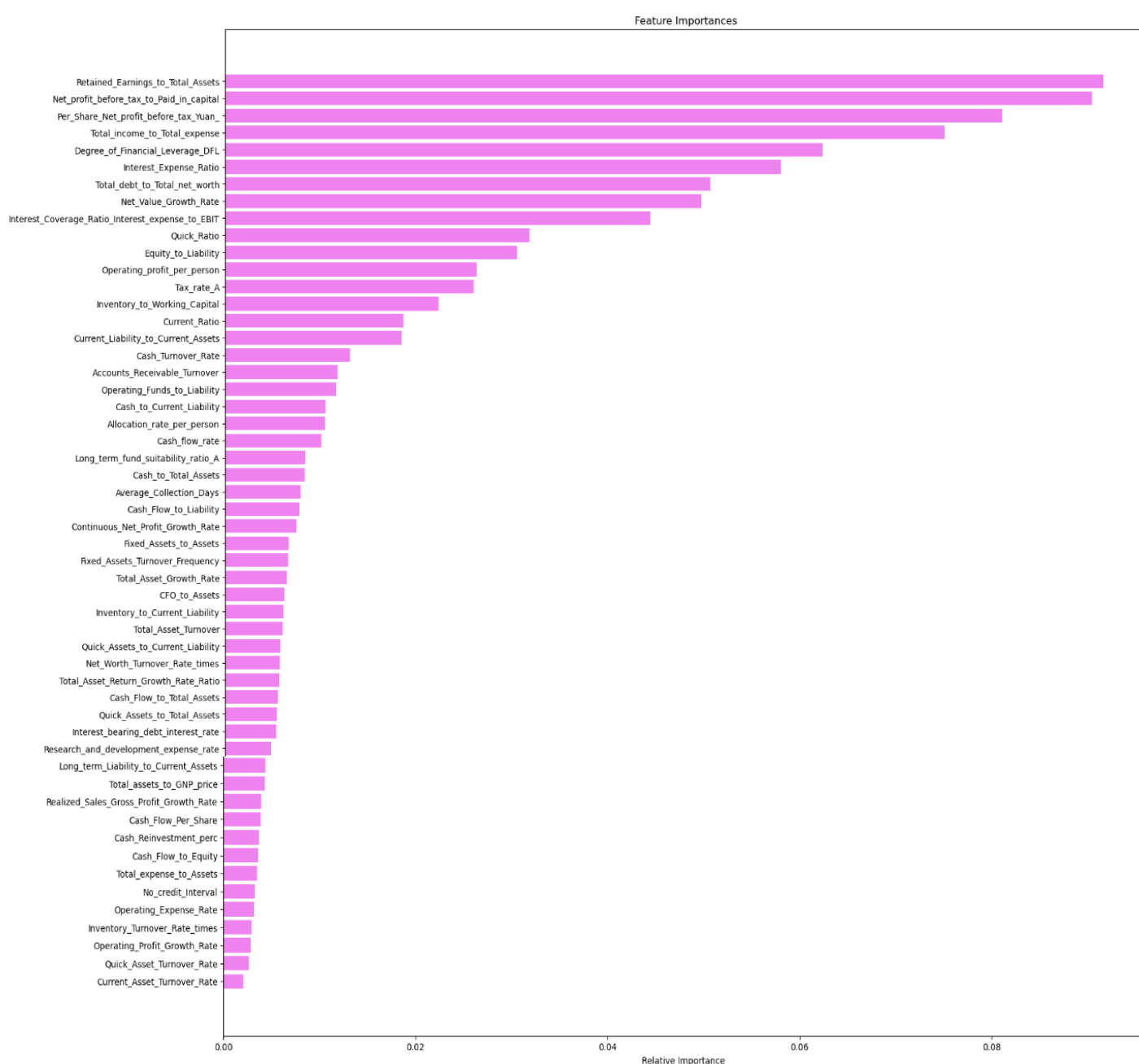


Figure 15- Feature Importance

6 - Actionable Insights & Recommendations:

Actionable Insights.

- From the above analysis we say that the dataset indicates a significant awareness of fraud risks among employees, but many still lack specific knowledge of reporting procedures and best practices.
- The all over analysis of the dataset reveals that data entry and processing stages are particularly vulnerable to fraud, highlighting the need for improved controls in these areas.
- We can say from the above analysis that the current fraud detection systems lack robustness, often depending on manual checks that result in slow responses to potential fraudulent activities.

Recommendations.

- Implement comprehensive training sessions focused on fraud awareness and reporting mechanisms to ensure all employees are equipped to identify and respond to suspicious activities.
- I recommend that, introduce automated controls in data entry and processing to minimize human error and reduce opportunities for fraud, including double verification and segregation of duties.
- I recommend that, invest in advanced fraud detection software that utilizes machine learning to identify anomalies in real-time, enabling quicker responses to potential fraud incidents.

PART B

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Problem Statement:

Context.

Investors face market risk, arising from asset price fluctuations due to economic events, geopolitical developments, and investor sentiment changes. Understanding and analysing this risk is crucial for informed decision-making and optimizing investment strategies.

Objective.

The objective of this analysis is to conduct Market Risk Analysis on a portfolio of Indian stocks using Python. It uses historical stock price data to understand market volatility and riskiness. Using statistical measures like mean and standard deviation, investors gain a deeper understanding of individual stocks' performance and portfolio variability. Through this analysis, investors can aim to achieve the following objectives:

- 1. Risk Assessment: Analysis historical volatility of individual stocks and the overall portfolio.
- 2. Portfolio Optimization: Use Market Risk Analysis insights to enhance risk-adjusted returns.
- 3. Performance Evaluation: Assess portfolio management strategies' effectiveness in mitigating market risk.
- 4. Portfolio Performance Monitoring: Monitor portfolio performance over time and adjust as market conditions and risk preferences change.

Overview of the Dataset.

	Date	Dish TV	Infosys	Hindustan Unilever	Vodafone Idea	Cipla
0	28-03-2016	86	608	867	67	514
1	04-04-2016	86	607	863	65	519
2	11-04-2016	85	583	853	66	506
3	18-04-2016	87	625	900	69	515
4	25-04-2016	89	606	880	71	532

Table1- Top 5 rows of the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  418 non-null   object
1   Dish TV               418 non-null   int64
2   Infosys               418 non-null   int64
3   Hindustan Unilever    418 non-null   int64
4   Vodafone Idea         418 non-null   int64
5   Cipla                 418 non-null   int64
dtypes: int64(5), object(1)
memory usage: 19.7+ KB
```

Table2- Basic Info of the dataset

	count	mean	std	min	25%	50%	75%	max
Dish TV	418.0	38.648325	31.944620	4.0	14.00	19.5	73.0	108.0
Infosys	418.0	1007.210526	455.089501	445.0	591.25	777.5	1454.0	1939.0
Hindustan Unilever	418.0	1906.344498	597.800173	788.0	1368.50	2083.0	2419.0	2798.0
Vodafone Idea	418.0	23.234450	20.264854	3.0	9.00	12.0	43.0	71.0
Cipla	418.0	756.614833	252.969619	370.0	556.00	637.0	946.0	1493.0

Table3- Statistical Summary of the dataset

```
Date          0
Dish TV       0
Infosys       0
Hindustan Unilever  0
Vodafone Idea  0
Cipla         0
dtype: int64
```

-Observations:

- There are 418 rows and 6 columns are present in the dataset.
- Dataset contain 5 int type and 1 object type data type, but here the object is basically datetime data type so we change the object datatype to datetime datatype.
- Hindustan Unilever have the highest mean value 1906.35 and highest max value 2798.0, Vodafone Idea lowest mean and max value.
- There are no missing values are present in the dataset.

1 - Stock Price Graph Analysis:

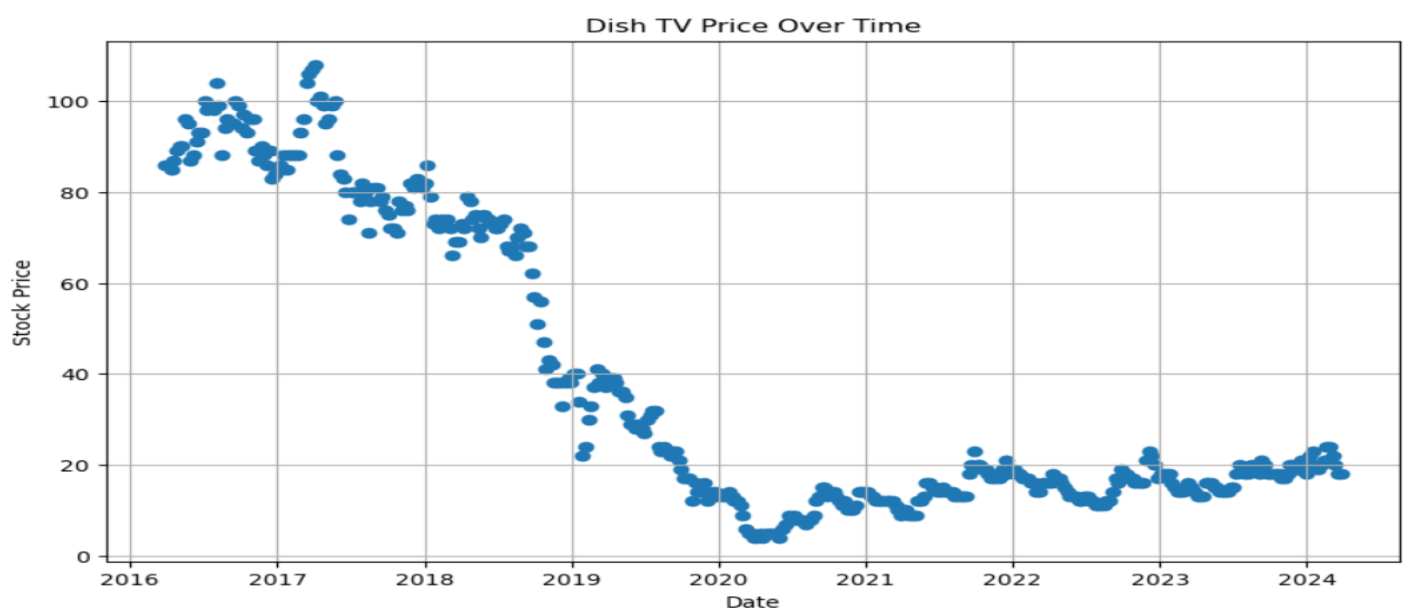


Figure 1- Dish Tv price over time

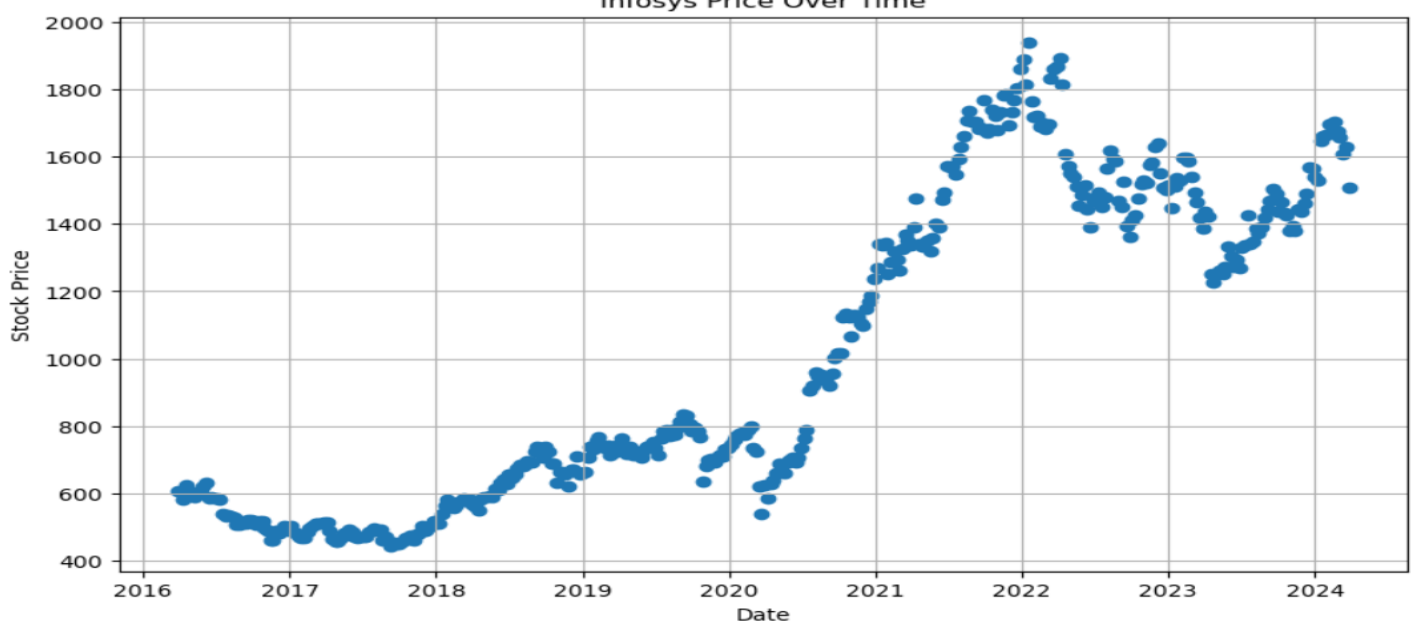


Figure 2- Infosys price over time

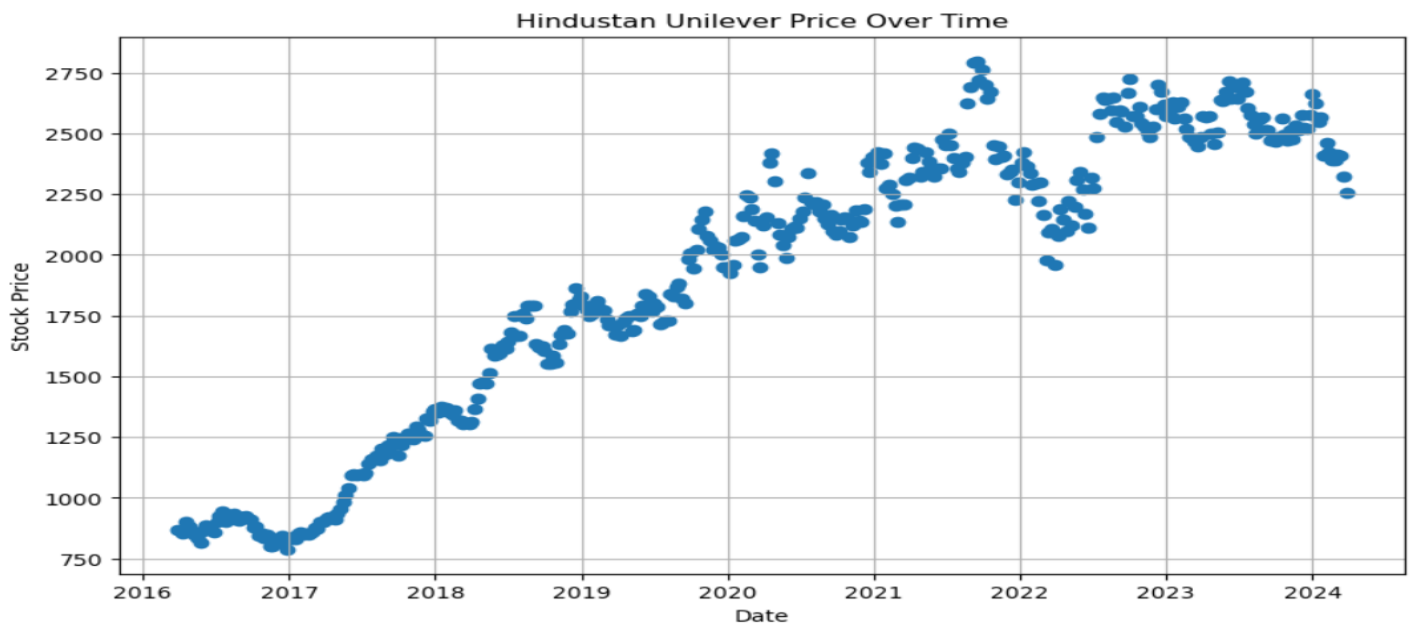


Figure 3- Hindustan Unilever price over time



Figure 4- Vodafone Idea price over time

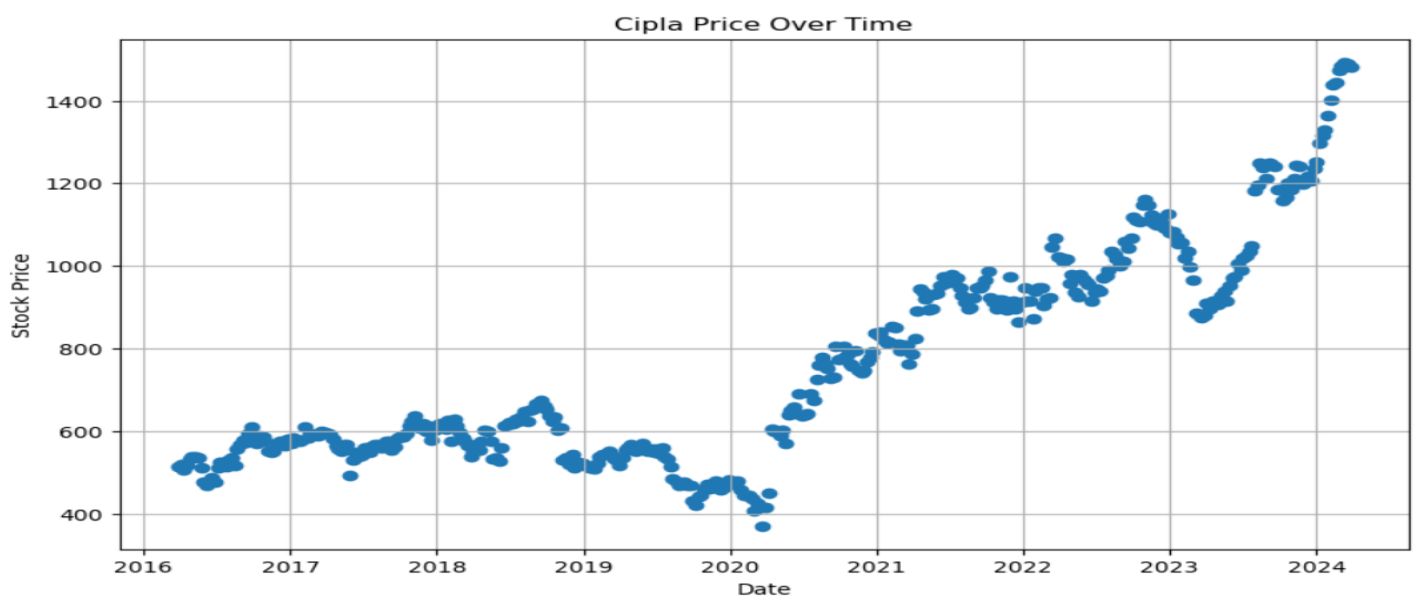


Figure 5- Cipla price over time

-Observations:

- The stock price of dish tv is continuously decreasing from the year 2016 to 2020 (around 100 to lower then 20rs) than it remains stable up to 2024.
- Stock price of Infosys is basically increasing over the time from the year 2016 to 2022 (around 600 to approx 2000rs) than it again fallen nearly 1200rs during the year 2023 to 2024
- The stock price of the Hindustan Unilever is continuously increasing over the time around 800 to 2700 rs.
- Price of Vodafone idea is also continuously decreasing from around 70 rs to bellow 10 rs and now currently it moves around nearly 20rs (year 2024).
- The stock price of Cipla is continuously increasing over the time period.

2 - Stock Returns Calculation and Analysis: Returns and Volatility Analysis.

> Return Calculation;

	Dish TV	Infosys	Hindustan Unilever	Vodafone Idea	Cipla
0	NaN	NaN	NaN	NaN	NaN
1	0.000000	-0.001646	-0.004624	-0.030305	0.009681
2	-0.011696	-0.040342	-0.011655	0.015267	-0.025367
3	0.023257	0.069564	0.053635	0.044452	0.017630
4	0.022728	-0.030872	-0.022473	0.028573	0.032477
5	0.011173	-0.001652	-0.014883	0.000000	0.009355
6	0.000000	-0.023412	-0.018627	-0.028573	0.001860
7	0.064539	0.023412	-0.021378	0.000000	-0.007463
8	-0.010471	-0.004971	-0.019395	-0.044452	-0.045985
9	-0.087969	0.031074	0.055934	0.029853	-0.066894
10	0.011429	0.017558	0.027399	-0.060625	-0.016914
11	0.033523	-0.072162	-0.023933	-0.031749	0.012712
12	0.021740	0.003396	0.009185	-0.016261	0.024949
13	0.000000	-0.006803	-0.019620	0.000000	-0.024949
14	0.072571	0.003407	0.047791	0.048009	0.073055
15	-0.020203	-0.006826	0.029559	-0.015748	0.027029
16	0.010152	-0.080185	0.018173	0.076373	-0.015355
17	-0.010152	-0.013072	-0.047731	-0.060625	-0.005820
18	0.059423	0.007491	0.032790	-0.015748	0.028765
19	-0.049271	-0.003738	-0.003231	0.015748	0.009407
20	-0.117783	-0.005634	0.008593	-0.115832	-0.034289
21	0.065958	-0.042314	-0.024907	0.000000	0.076458
22	0.021053	0.000000	-0.006601	-0.017700	0.019556
23	-0.010471	0.013659	0.016421	-0.093526	0.017452
24	0.000000	-0.005831	0.004334	-0.019803	-0.010435
25	0.051293	0.019306	-0.010870	0.019803	0.039422
26	-0.010050	-0.003831	-0.006579	-0.019803	0.023257
27	-0.051825	-0.005775	-0.034699	-0.040822	-0.040206
28	0.031416	-0.019494	0.002275	0.020619	-0.025975
29	-0.042111	0.017561	-0.041769	-0.085158	0.032790
30	0.031749	0.003861	0.009434	0.043485	-0.006814
31	0.000000	-0.039298	-0.018958	-0.021506	-0.017242
32	-0.075712	-0.014127	0.016608	-0.021979	-0.044452
33	-0.022728	-0.065081	-0.058128	-0.068993	-0.003643
34	0.033902	0.004329	0.001246	0.023530	0.003643
35	-0.022473	0.056678	0.033071	0.022990	0.033962
36	-0.022990	-0.014389	0.003608	0.000000	0.008749
37	0.034289	0.016427	0.014303	0.044452	0.003478
38	-0.069796	0.024146	-0.033698	0.000000	-0.019282
39	0.011976	-0.018055	-0.036141	-0.067441	0.027925
40	0.011834	0.024001	0.049515	0.045462	-0.019114
41	0.011696	-0.040328	0.012005	-0.022473	0.024265

Table 4- Return Calculations

-Observations:

- From the return calculation we see a strong positive change in Cipla's performance, suggesting a potential increase in market confidence or a successful product launch.
- We see a significant negative change for Vodafone Idea, indicating potential issues affecting its market position or financial stability.

>Average Returns;

```
Vodafone Idea      -0.003932
Dish TV            -0.003751
Infosys            0.002180
Hindustan Unilever 0.002294
Cipla              0.002538
dtype: float64
```

-Observations:

- Cipla shows a positive change, indicating potential growth or improved market performance.
- Vodafone Idea is facing a small decline, which might be due to competition or operational issues. This situation suggests the need for a closer look at the company's strategies and market conditions to identify risks and inform stakeholders about possible corrective actions.

> Volatility;

```
Hindustan Unilever 0.028845
Infosys            0.036102
Cipla              0.036759
Dish TV            0.091333
Vodafone Idea      0.113747
dtype: float64
```

-Observations:

- Vodafone Idea has the highest positive change among the companies. This strong growth suggests that the company may have overcome past challenges or taken advantage of new market opportunities.
- Dish TV also exhibits a strong positive change, indicating potential recovery or growth in its market position.

> Visualizing Returns and Volatility;

	Mean	Volatility
Dish TV	-0.003751	0.091333
Infosys	0.002180	0.036102
Hindustan Unilever	0.002294	0.028845
Vodafone Idea	-0.003932	0.113747
Cipla	0.002538	0.036759

Observations:

- We see that the both Vodafone Idea and Dish TV exhibit high volatility, which is indicate that a significant price fluctuation. This could suggest that these companies are facing market uncertainties or challenges that may impact their stock performance.
- We see that Cipla and Hindustan Unilever show positive mean returns, which is suggest a stable performance and potential growth. Their lower volatility compared to Vodafone Idea and Dish TV may indicate more consistent performance, making them attractive options for lower Risk investors.

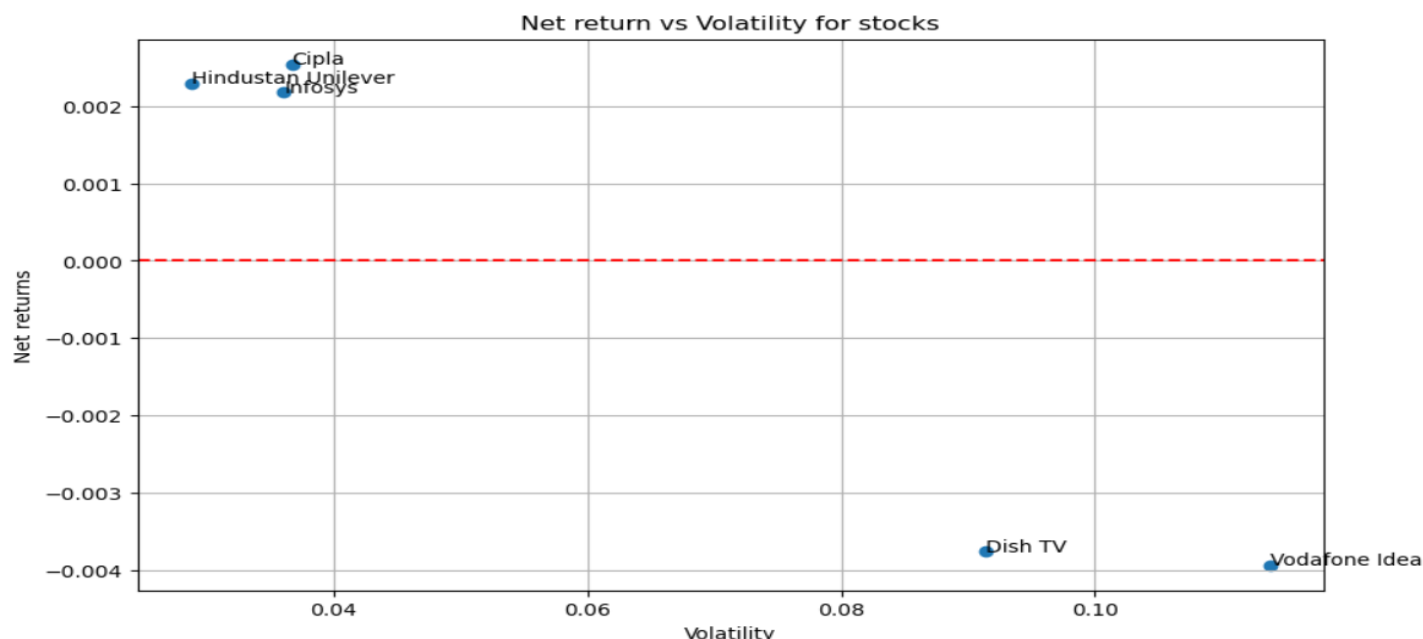


Figure 6- Net return vs volatility for stocks

Observations:

- The scatter plot shows a trade-off between risk and returns. Stocks like Vodafone Idea and Dish TV have high volatility but lower returns, indicating they carry more risk. Investors should consider their risk tolerance when evaluating these stocks.
- Cipla and Hindustan Unilever show low volatility and positive returns, indicating stable performance. This makes them attractive for risk-averse investors, offering safer investments with moderate growth potential compared to more volatile stocks.

3 - Actionable Insights & Recommendations:

Actionable Insights.

- We observe from the whole analysis and see that the companies like Cipla and Hindustan Unilever exhibit lower volatility with positive returns, suggesting that investors seeking stability should consider these stocks for their portfolios.
- The high volatility of Vodafone Idea and Dish TV indicates potential for higher returns but also greater risk. Investors should assess their risk tolerance before investing in these stocks.

Recommendations.

- To balance risk, investors should diversify their portfolios by including both stable stocks like Cipla and Hindustan Unilever and higher-risk stocks like Vodafone Idea and Dish TV. This strategy can help mitigate potential losses while capturing growth opportunities.
- Regularly review market conditions and company performance, especially for more volatile stocks. Staying informed can help investors make timely decisions to adjust their portfolios in response to changing market dynamics.